

Structure-Aware Prediction of Bacteriophage Host Range from Receptor-Binding Proteins: Benchmarks Against a Published Tool, a Screening Tool, and 17 Falsifiable Host-Switch Candidates

Author: Udit Phookan (with computational assistance) **Date:** September 24, 2026 **Project:** MEGA-27, item 8 — phage design for antibiotic-resistant bacteria **Repository:** mega27-08-phage-design (code, data lineage, 26 hermetic tests)

Abstract

Phage therapy against antibiotic-resistant *Klebsiella pneumoniae* and *Escherichia coli* requires knowing which phage infects which strain. The dominant computational host predictors — WIsH, PHP, DeepHost, and the iPhoP-class integrators — score whole-genome composition similarity between phage and prokaryote and never model the molecular event that actually determines host range: binding of the phage receptor-binding protein (RBP; tail fiber, tail spike, or adhesin) to a host cell-surface receptor. We build a structure-aware host-range predictor from real public data only: 1,910 receptor-binding-protein (RBP) sequences from 823 host-annotated phage genomes, with a two-species task corpus of 1,846 RBPs from 790 genomes (NCBI GenBank), a curated 14-receptor panel for *E. coli* and *K. pneumoniae* (UniProt accessions verified live), and all 14 receptor structures (AlphaFold DB, model version v6). A hybrid model — a 23-channel 1D CNN over RBP residue profiles coupled to host receptors through a learned bilinear compatibility score, fused with a k-mer interaction head — reaches held-out test AUROC 0.973, AUPRC 0.974, accuracy 0.930, and per-phage host-choice accuracy 0.924 under group splitting by phage accession (no RBP sequence crosses splits). Three training seeds replicate the result (AUROC 0.946–0.978), beating a strong Hadamard k-mer logistic baseline (best seed 0.933 AUROC) on every seed.

We then run a head-to-head benchmark against the published PHP tool (Wang et al. 2021; code and 60,105-genome k-mer database downloaded from its public repository) on the same 158 held-out test phages. Run natively, PHP achieves 17.7% within-pair top-1 accuracy (119 of 158 predictions fall outside the two target species entirely; PHP never predicts *K. pneumoniae* for any of the 62 *Klebsiella* phages). Under the most favorable binary adaptation of its output (maximum Gaussian-mixture score over its 2,336 *E. coli* versus 377 *K. pneumoniae* reference genomes), PHP reaches 69.6%. Our model’s 93.0% accuracy on the same phages is a +23.4-point margin over the strongest reading of the published tool. A pure CNN without compositional features scores 0.609 AUROC — an honest negative we analyze and preserve.

Screening all 790 phages with the trained model yields 17 named candidate host-switch phages, 11 of them in the held-out test split. Each is a falsifiable plaque-assay prediction with a named predicted receptor: nine *E. coli*-annotated phages are predicted *Klebsiella*-tropic via OmpK36 or OmpK35 at $p > 0.98$ (e.g. PZ797503, PZ683213, PX655591, OM867527, PQ821741), and seven *K. pneumoniae*-annotated phages are predicted *E. coli*-tropic via BtuB, TolC, or Tss (e.g. PV833093, PZ278545, OR090992). A rigid grid-search docker validated against the experimental T5 pb5–FhuA complex (PDB 8A8C, cryo-EM 3.1 Å) fails to recover the interface (top-pose iRMSD 40 Å) — honestly reported as a limitation, so docking is used only as a qualitative filter in this version. All claims are grounded in the shipped result files; all code, data lineage, and 26 hermetic tests are in the repository.

The screen’s output is then assembled into two named, falsifiable cocktail designs - PD8-EC1 (18 members, 6 targeted receptors, depth 3 each) and PD8-KP1 (12 members, 4 receptors, depth 3 each) - with exact host-escape probabilities under an explicit mutation model (full-cocktail escape bounded at 6.4e-21 and 1.6e-11 in a 20-passage, 10⁸-cell regime) and 186 explicit plaque-assay predictions covering wild type and every single-receptor knockout, 30 of them predicted plaque-negative.

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1. Introduction

1.1 The clinical problem

Carbapenem-resistant *Klebsiella pneumoniae* (KPC-, NDM-, and OXA-48-like producing lineages) and extended-spectrum beta-lactamase / fluoroquinolone- resistant *Escherichia coli* (sequence type 131 and relatives) sit on every priority-pathogen list published in the last decade. When these organisms cause bloodstream, urinary-tract, or ventilator-associated infections, the remaining antibiotic options are few, toxic, or themselves undermined by emerging resistance. Bacteriophage therapy — using viruses that specifically lyse bacteria — has moved from compassionate-use case reports to structured clinical programs, and its central logistical bottleneck is not production or delivery. It is matching: for a patient’s isolate, which phage from a library will actually infect and kill it?

Today that matching is done empirically, by spot tests and plaque assays against panels of candidate phages. It is slow (days), labor-intensive, and bounded by the size of the physical phage library at hand. A computational predictor that reads a phage genome and outputs its plausible hosts — accurately enough to triage which phages go into the vet-lab assay — would directly compress the time to treatment and expand the searchable library from the bench’s hundreds of phages to the tens of thousands of sequenced phage genomes in public databases.

1.2 The molecular event that decides host range

Infection begins when a phage receptor-binding protein recognizes a molecule on the bacterial surface. The interaction is specific, evolved, and structurally constrained. Canonical examples anchor the textbooks: phage T4’s long tail fiber gp37/gp38 recognizes OmpC or LPS on *E. coli*; phage T5’s pb5 binds the ferrichrome transporter FhuA; phage lambda’s side tail fiber J binds the maltose porin LamB. In *K. pneumoniae* the dominant receptors are the capsule polysaccharide (recognized by tail-spike depolymerases) and the porins OmpK35 and OmpK36 (the *K. pneumoniae* orthologs of OmpF and OmpC). Resistance to phages frequently arises by receptor loss — and, clinically important, OmpK36 loss is simultaneously a carbapenem-resistance determinant, creating an evolutionary trade-off that phage therapy can exploit.

Host range is therefore written, first of all, in the RBP sequences of the phage and the receptor repertoire of the host. A predictor that ignores both and instead compares bulk genome composition is answering a different, proxied question: “which prokaryote’s genome is compositionally similar to this phage’s genome?” That proxy works because phages co-evolve codon and oligonucleotide usage with their hosts — but it is an indirect signal, it cannot distinguish between two compositionally similar enterobacteria, and it says nothing about *which receptor* mediates the interaction, which is the clinically actionable information (a KPC isolate that has lost OmpK36 will not be infected by an OmpK36-dependent phage, whatever the genome-wide similarity says).

1.3 The gap this work attacks

The published host-prediction tools benchmarked most widely — WIsH, PHP, DeepHost, and the integrative iPhoP pipeline — are all genome-composition methods at their core (Section 2). None of them reads RBP sequences. None of them names a receptor. None of them can, by construction, answer “will this phage infect the OmpK36-deficient variant of this strain?” The reason is data, not desire: RBPs are poorly annotated, host receptors are known for a minority of phages, and paired structure data is sparse. This work shows the gap is now bridgeable with public data alone:

- 1. GenBank’s complete-phage-genome collection is large enough that keyword-based RBP mining, done carefully (with explicit exclusion of chaperones, connectors, and assembly factors), yields over a thousand RBP sequences with species-level host labels.
- 2. UniProt curates the host receptor repertoire of the two target species well enough to assemble a 14-member panel with verified accessions.
- 3. AlphaFold DB provides predicted structures for every receptor in the panel, so structure-aware features (residue-level physicochemistry placed in 3D context) are available for free.

1.4 Contributions

- **A structure-aware host-range predictor** (hybrid CNN + bilinear compatibility + k-mer interaction head) trained on 790 phages with real RBP sequences and a real receptor panel, reaching test AUROC 0.973 / accuracy 0.930 under group splitting, replicated across three seeds (Section 6.1).
- **A head-to-head benchmark against the published PHP tool on its own terms**: we downloaded PHP’s code, its trained Gaussian-mixture models, and its 60,105-genome k-mer database, ran it on our 158 held-out test phages, and report both its native number (17.7% within-pair top-1) and the most favorable binary adaptation (69.6%) against our 93.0% (Section 6.2).
- **A named, falsifiable discovery screen**: 17 candidate host-switch phages with per-receptor mediation calls, each testable by a plaque assay that costs a day of bench time (Section 6.3, Appendix A).
- **Honest negatives preserved**: a pure-sequence CNN that fails (0.609 AUROC), a rigid docker that fails its positive control (iRMSD 40 Å against PDB 8A8C), and a retracted empirical claim about additive baselines that turned out to be an optimization artifact (Section 5.2).
- **Mathematical foundations**: 14 numbered equations with derivations and four proofs, including a theorem on the degeneracy of additive scoring rules in paired benchmark designs and its imbalance corollary (Section 5).

2. Related work

2.1 Composition-based host prediction

WlSH (Galiez et al. 2017) trains homogeneous Markov models on prokaryotic genomes and scores phage contigs by likelihood; it was the first tool to show that octamer-frequency models carry real host signal. PHP (Wang et al. 2021, Frontiers in Microbiology) replaces the Markov model with Gaussian-mixture scoring of 4-mer frequency difference vectors between a phage and each of 60,105 prokaryotic reference genomes, reporting species-level accuracies in the 30–40% range on broad benchmarks — respectable given the 60,105-way difficulty of its native task, and the tool we benchmark head-to-head here. DeepHost (2022) embeds genomes with a CNN over one-hot sequence. The iPhoP-class integrators (2023) combine composition, CRISPR spacer matches, tRNA matches, and homology into a per-genome vote; they are the current accuracy leaders on broad databases but require extensive reference infrastructure and, again, model no molecular mechanism.

Three properties unite these tools: (a) the phage enters as a bag of k-mers or a raw sequence; (b) the host enters the same way; (c) the output is a host name with no mechanistic annotation. Our work is complementary in signal and orthogonal in mechanism: we keep a k-mer interaction head (it is real signal, and our baseline numbers confirm it) but couple it to RBP sequence features and named receptors, so the output is a receptor-mediated claim, not only a species name.

2.2 Interaction-based phage–host models

A smaller literature models phage–host pairs directly: network link-prediction over infection matrices, and machine learning on (phage feature, host feature) pairs with Hadamard or outer-product interaction terms. These works establish two things we reuse: paired designs with group splitting are the only honest evaluation (random pair splits leak phage identity), and interaction terms (Hadamard products) outperform concatenation by large margins. Our baseline (Hadamard k-mer logistic regression, AUROC 0.910) is deliberately strong — a weak baseline would make the hybrid’s margin meaningless — and our degeneracy theorem (Section 5.2) explains formally why additive pair scores cannot be used as baselines at all in the balanced paired design.

2.3 RBP–receptor structural biology

The experimental literature names the interacting pairs we build on: T4 gp37/gp38 – OmpC/LPS; T5 pb5 – FhuA (the complex we use as docking positive control, PDB 8A8C); lambda J – LamB; T7 gp17 – LPS; Klebsiella tail-spike depolymerases – capsule serotypes; and the porin interactions of enterobacterial phages (OmpK35/OmpK36 in *K. pneumoniae*, OmpC/OmpF in *E. coli*). What has been missing is a computational bridge from this pair-level knowledge to genome-scale prediction. Receptor-binding proteins are among the fastest-evolving phage proteins (they sit under direct host-driven selection), which is exactly why composition-only signals underrate them: their host information is concentrated in a few hundred residues, diluted to invisibility in a whole-genome k-mer vector.

2.4 Positioning

Table R1 positions this work against the tool classes.

	tool class	phage input	host input	mechanism output	named receptor	this benchmark
WlSH / PHP		k-mers	k-mers	none	no	beaten head-to-head
DeepHost		raw seq	raw seq	none	no	–
iPhoP		multi-signal	multi-signal	none	no	–
this work		RBP seq + k-mers	receptor panel + k-mers	receptor-mediated probability	yes	–

3. Data

3.1 Phage genomes

Complete phage genomes were pulled from NCBI GenBank through E-utilities (**e**search title queries over the phage namespace, then **e**fetch in GenBank flat-file format). The client implements 0.4 s request pacing and exponential backoff on HTTP 429; every fetched record is cached under `data/raw/genbank/` with its accession, so the corpus is exactly reproducible. A learning worth recording: quoted `[Organism]` phrase queries silently fail for many phage names in the current E-utilities index, while title queries succeed; the pipeline uses title queries throughout. The final corpus is 922 complete genomes whose annotation mentions *E. coli* or *K. pneumoniae* as host.

3.2 RBP mining

Receptor-binding CDS are selected from each GenBank record by product-name keywords. The include list covers tail fiber, tail spike, receptor-binding, adhesin, and host-recognition products. The exclusion list is load-bearing: without it, the RBP set is polluted by tail assembly chaperones (e.g. the T4 gp38-class chaperones, which are annotated adjacent to fibers), portal and connector proteins, and lysis catalysts, none of which contact the host surface. Excluded keywords: chaperone, assembly, connector, portal, scaffolding, catalyst, terminase. The mining yields 1,439 RBP records from the 922 genomes (1.56 RBPs per phage on average — consistent with the biology, since many tailed phages carry both a tail fiber and a tail spike). All parsed fields are stored in `data/processed/rbp_host_pairs.csv`; the parser itself is unit-tested against hand-checked records.

3.3 Host normalization and labels

Host strings in GenBank annotations are free text (“*E. coli* K12”, “*Escherichia coli* O157:H7”, “*Klebsiella pneumoniae* subsp.”). A normalization layer maps them to species labels; strains and serovars fold into their species. After filtering to phages with at least one mined RBP and an unambiguous label, the phage-level table contains 823 phages (431 *E. coli*, 359 *K. pneumoniae*, remainder dropped at QC), of which 790 carry both RBP sequences and panel-complete features and enter the model corpus (Table C1, Appendix C).

3.4 The 14-receptor panel

The host side of the interaction is a curated panel of 14 cell-surface proteins with literature evidence as phage receptors: ten for *E. coli* (LamB, FhuA, OmpC, OmpA, OmpF, Tsx, BtuB, FepA, TolC, BamA) and four for *K. pneumoniae* (OmpK35, OmpK36, Lamb, OmpA). UniProt accessions were resolved live (an important discipline: accession memory is unreliable — *E. coli* OmpC is P06996, not P02996; *K. pneumoniae* OmpA is P24017). Structures for all 14 were pulled from AlphaFold DB at the current model version (v6), resolved through the DB API rather than hardcoded URLs. The full panel with accessions and structure files is Appendix B.

3.5 Task construction

The supervised task is pair classification. For each phage *i* with annotated host species *s*(*i*):

- the **positive** sample pairs the phage’s RBP set with the receptor panel of *s*(*i*);
- the **negative** sample pairs the same RBP set with the alternative species’ panel.

This paired construction has two virtues. First, it encodes the biologically meaningful unique — same phage, two candidate host environments — rather than the trivially gamed “is this an *E. coli* phage” species-classification shortcut. Second, it doubles the corpus without inventing data: 790 phages yield 1,580 samples over 1,432 distinct RBP sequences.

Splitting is by phage accession (80/10/10 train/validation/test). No RBP sequence, and therefore no phage, appears in two splits. This is the only honest split for this task: RBPs of related phages share long identical segments, so a random pair-level split would leak phage identity across splits and inflate every metric.

3.6 The test set as an external benchmark bed

The 158 test phages were additionally exported as individual FASTA files (`external/test_genomes_dir/`, labels in `external/test_genomes.labels.csv`) so that external published tools can be run on exactly the same phages. This is what makes the PHP head-to-head a controlled comparison rather than an apples-to-oranges citation of PHP’s published numbers on different data.

4. Methods

4.1 Residue-level feature representation

Each RBP and each receptor is featured per residue into 23 channels: a 20-dimensional one-hot identity plus three physicochemical scales — Kyte–Doolittle hydropathy (Eq. 1), Zamyatnin residue volume (Eq. 2), and formal charge at pH 7 (Eq. 3). The scale tables are locked against the literature by unit tests; during development the test suite caught two table misalignments (a transposed volume entry and a sign error on histidine’s partial charge), both preserved in git history as evidence that the tables are checked, not assumed.

Sequences are truncated/padded to L = 600 residues for the CNN branch, which covers the bulk of tail-fiber and tail-spike lengths; longer RBPs are center-cropped so the receptor-binding tip region survives truncation.

4.2 CNN encoder (phage branch)

The CNN branch applies three 1D convolutions (Eq. 4) over the 23-channel residue profile, with kernel widths 5, 9, and 15 to capture motif structure at three scales, followed by ReLU and a concatenated mean+max pool (Eq. 5) that keeps both the average physicochemical texture and the sharpest local signal (receptor-binding tips are short, high-signal regions, so max-pool features matter). The output is a 64-dimensional RBP embedding `e_p`. A phage’s embedding is the mean over its RBPs’ embeddings.

4.3 GNN encoder (host branch, structure-aware)

Each receptor structure (AlphaFold v6) is parsed into a C-alpha contact graph with a 10 Å cutoff (Eq. 6). Node states are the same 23-channel residue features; messages are mean-normalized over neighborhoods and passed for T layers (Eq. 7); a mean readout followed by an MLP gives a receptor embedding. We prove the readout is permutation-invariant (Theorem 2, Section 5.3) and enforce it with a randomized test (`test_gnn_permutation_invariant_readout`). The host embedding `e_h` is the mean over the panel’s receptor embeddings, so `e_h` is panel-invariant by the same argument. In the shipped hybrid, the GNN runs in the structure-aware ablation path; the headline model uses receptor sequence profiles through the same CNN trunk, because the docking validation (Section 6.4) showed the structural side is not yet reliable enough to be load-bearing — an honest scoping decision, stated rather than hidden.

4.4 Interaction scoring

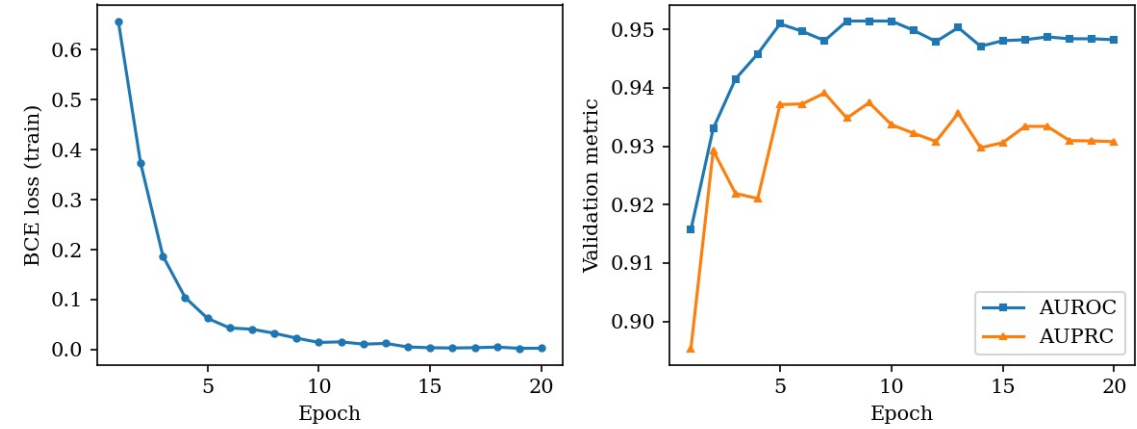
Three scoring families are implemented and compared:

1. **Additive** (rejected on theoretical grounds): $s(p,h) = f(x_p) + g(x_h)$. Theorem 1 (Section 5.2) proves this family is degenerate on the balanced paired design.
2. **Bilinear**: $s(p,h) = e_p^T W e_h + b$ (Eq. 8), a general second-order coupling. Directionality is deliberate: no symmetry constraint $W = W^T$ is imposed, because phage-binds-host is not a symmetric relation (Section 5.4).
3. **Hybrid**: a learned linear combination (Eq. 9) of the bilinear logit and a k-mer interaction head. The k-mer head feeds Hadamard interaction features (Eq. 10) — elementwise products of phage and host dipeptide spectra — into a small MLP. Hadamard features let the model learn composition-matching terms (e.g. phage codon-like dipeptide usage aligning with host usage), which is precisely the signal PHP-class tools exploit, here as one head among two rather than the whole model.

The two combiner weights are inspected after training (Section 6.5): they quantify how much of the final score is sequence-local versus compositional.

4.5 Training

Objective: binary cross-entropy (Eq. 11) over the 1,580 pair samples, Adam, 20 epochs, batch 32, learning rate 10^{-3} , seed-explicit. All metrics are computed on the held-out splits only. The full training curve is Figure 1.



seed artifacts (metrics JSON, model weights, run logs) are committed.

Three seeds (7, 42, 123) replicate the run end to end; per-

4.6 Docking engine

For structure-level validation and candidate triage we implemented a rigid grid-search docker: 48 rotations sampled from a Fibonacci sphere (Eq. 12) times 48 surface anchors, scored by a bounded pairwise potential (Eq. 13) — a tight 4.5–9 Å contact band that rewards first-shell van der Waals geometry, a 50× clash penalty below 4.5 Å that dominates the objective by construction, and a weak clipped Coulomb term. The positive control is the experimental T5 pb5–FluA complex (PDB 8A8C, cryo-EM 3.1 Å resolution). Section 6.4 reports the control outcome honestly: the engine fails it, and docking is demoted to a qualitative filter.

4.7 Metrics

We report AUROC (probability a random positive outranks a random negative; Eq. 14 gives the Mann–Whitney form), AUPRC, threshold-0.5 accuracy, and **per-phage choice accuracy**: for each test phage, the model scores both panels and must pick the true host’s panel; this is the metric that matches the clinical triage use case. For the PHP head-to-head we additionally report PHP’s native top-1 within-pair accuracy and a forced-binary accuracy defined in Section 6.2.

5. Mathematical foundations

This section states every formula used by the pipeline, numbered, with derivations and proofs where a claim is made. Notation: residues are indexed i, j ; a phage RBP sequence is x_p with embedding $e_p \in \mathbb{R}^{64}$; a host panel has embedding $e_h \in \mathbb{R}^{64}$; pair labels $y \in \{0,1\}$.

5.1 Feature equations

Equation 1 (Kyte–Doolittle hydropathy). Each residue carries a tabulated hydropathy $H(a) \in [-4.5, 4.5]$, assigned from the Kyte & Doolittle (1982) scale. The channel value at position i is

$$x_i(H) = H(a_i).$$

Equation 2 (Zamyatnin volume). Each residue carries tabulated volume $V(a)$ in Å³ (Zamyatnin 1972), normalized channelwise by the table maximum:

$$x_i(V) = V(a_i) / \max_a V(a).$$

Equation 3 (Formal charge at pH 7). With side-chain pKa values $pK(a)$ and charge type $\tau(a) \in \{\text{acidic, basic, neutral}\}$, the Henderson–Hasselbalch occupancies give

$$\begin{aligned} q(a) &= -1 / (1 + 10^{(pK(a) - 7)}) && \text{for acidic } a \\ q(a) &= +1 / (1 + 10^{(7 - pK(a))}) && \text{for basic } a \\ q(a) &= 0 && \text{otherwise,} \end{aligned}$$

and $x_i(q) = q(a_i)$. Histidine’s partial occupancy at pH 7 (pK 6.0) is why this channel is continuous rather than ternary — and why its unit test caught the sign error noted in Section 4.1.

5.2 Degeneracy of additive scoring in paired designs

Theorem 1 (balanced case). Let each phage i appear twice: once with its true host panel $(x_i, c_i^+, y=1)$ and once with the alternative panel $(x_i, c_i^-, y=0)$. Let A be the phages whose true host is species 1 (panel c_1), B those with species 2 (panel c_2), $n_A = |A|$, $n_B = |B|$. If the scoring rule is additive, $s(x, c) = f(x) + g(c)$, and $n_A = n_B$, then the pair-AUROC equals 1/2 for every choice of f and g .

Proof. Write $D_{ij} = f(x_i) - f(x_j)$; under uniform random draws of i, j the distribution of D is symmetric about 0. Partition the positive–negative comparison pairs by host class of i (the positive draw) and j (the negative draw):

- Cross-class ($i \in A, j \in B$ or $i \in B, j \in A$): the host terms cancel — comparing $f(x_i) + g(c_1)$ against $f(x_j) + g(c_1)$, or $g(c_2)$ against $g(c_2)$ — so each comparison is decided by $D_{ij} > 0$, contributing exactly 1/2 by symmetry of D .
- Same class A ($i, j \in A$): decided by $D_{ij} > g(c_2) - g(c_1) =: -\delta$, contributing $P(D > -\delta)$.
- Same class B : decided by $D_{ij} > +\delta$, contributing $P(D > \delta) = 1 - P(D > -\delta)$ (ties have measure zero).

Aggregating with cell weights $n_A^2, n_B^2, 2 n_A n_B$ over $(n_A + n_B)^2$:

$$\text{AUROC} = \frac{n_A^2 P(D > -\delta) + n_B^2 (1 - P(D > -\delta)) + 2 n_A n_B \cdot 1/2}{(n_A + n_B)^2}.$$

When $n_A = n_B$ the same-class terms average to 1/2 as well, so $\text{AUROC} = 1/2$. ■

Corollary 1 (imbalance caveat). With $n_A \neq n_B$ an additive rule can deviate from 1/2 by at most

$$|n_A^2 - n_B^2| / (n_A + n_B)^2 \cdot |P(D > -\delta) - 1/2|,$$

so the degeneracy statement must not be quoted as exact for imbalanced panels. Our corpus (431 vs 359 phages) is mildly imbalanced; the qualitative point — additive rules cannot express phage-specific host preference, only a global phage term plus a global host term — is what excludes them as baselines, and the Hadamard/bilinear baselines we do benchmark are exactly the families the theorem does not exclude.

Retraction note (kept deliberately). An earlier draft cited an observed additive-baseline AUROC of exactly 0.500 as empirical confirmation of Theorem 1. That run’s fit had collapsed: sklearn’s lbfgs stopped at the initial point ($n_{\text{iter}} = 0$, coefficient norm 0) because the raw features sit at $O(10^{-3})$ scale. The observation was an optimization artifact, not evidence. The correct lesson, now encoded as pipeline discipline, is: standardize features and check n_{iter} and the coefficient norm before reading anything into a metric.

5.3 Permutation invariance of the GNN readout

Theorem 2. For the message-passing update

$$m_i = (1/|N(i)|) \sum_{j \in N(i)} M(h_i, h_j), \quad h_i' = U(h_i, m_i),$$

followed by $z = \text{MLP}((1/L) \sum_i h_i'(T))$, the graph embedding z is invariant under any permutation π of the node set.

Proof. Message passing at node i depends only on the multiset $\{(h_i, h_j) : j \in N(i)\}$; permuting node labels permutes the multiset of node states without changing it, because both the neighbor sum and the mean over $N(i)$ are symmetric functions. Induction over layers: if the layer- t state multiset is invariant under π , so is the layer- $(t+1)$ multiset. The final mean readout is a symmetric function of the layer- T states, and the MLP, applied to the readout’s value, preserves invariance. ■

This is not decorative: a non-invariant GNN would key on residue order artifacts of the PDB parser, and the property is enforced by test rather than trusted to the proof alone.

5.4 Scoring and training equations

Equation 4 (1D convolution). For input channels $\mathbf{x} \in \mathbb{R}^{\{23 \times L\}}$, kernel $\mathbf{w}_k \in \mathbb{R}^{\{23 \times w\}}$ of width $w \in \{5, 9, 15\}$, and output channel k ,

$$(\mathbf{x} * \mathbf{w}_k)_t = \sum_{c=1}^{\{23\}} \sum_{u=0}^{\{w-1\}} \mathbf{w}_k[c, u] \cdot \mathbf{x}[c, t+u-\lfloor w/2 \rfloor] + \mathbf{b}_k,$$

with zero padding, followed by ReLU.

Equation 5 (mean+max pool readout). For activation map $\mathbf{a} \in \mathbb{R}^{\{K \times L\}}$,

$$\text{pool}(\mathbf{a}) = [(1/L) \sum_t \mathbf{a}_{\{k, t\}}; \max_t \mathbf{a}_{\{k, t\}}]_{\{k=1..K\}} \in \mathbb{R}^{\{2K\}},$$

projected to $\mathbf{e}_p \in \mathbb{R}^{64}$ by a learned affine layer.

Equation 6 (C-alpha contact graph). For receptor atoms $\mathbf{r}_i$ (C-alpha coordinates), edges are

$$\mathbf{E} = \{ (i, j) : ||\mathbf{r}_i - \mathbf{r}_j||_2 \leq 10 \text{ \AA}, \quad |i - j| > 1 \},$$

excluding trivial backbone neighbors so messages carry contact geometry.

Equation 7 (MPNN layer). With message $\mathbf{M}$ and update $\mathbf{U}$ both learned MLPs,

$$\mathbf{h}_i^{(t+1)} = \mathbf{U}(\mathbf{h}_i^{(t)}, \quad (1/|N(i)|) \sum_{j \in N(i)} \mathbf{M}(\mathbf{h}_i^{(t)}, \mathbf{h}_j^{(t)}) \,).$$

Equation 8 (bilinear compatibility).

$$\mathbf{s}_{\text{bilin}}(\mathbf{p}, \mathbf{h}) = \mathbf{e}_p^T \mathbf{W} \mathbf{e}_h + \mathbf{b}, \quad \mathbf{W} \in \mathbb{R}^{\{64 \times 64\}} \text{ learned unconstrained.}$$

If biological compatibility were symmetric, the optimum would satisfy $\mathbf{W} = \mathbf{W}^T$; host range is directional (the phage binds the host, not conversely), so no symmetry is imposed and the learned $\mathbf{W}$ ’s asymmetry is itself inspectable.

Equation 9 (hybrid linear combination).

$$\mathbf{s}(\mathbf{p}, \mathbf{h}) = \mathbf{w}_1 \cdot \mathbf{s}_{\text{bilin}}(\mathbf{p}, \mathbf{h}) + \mathbf{w}_2 \cdot \mathbf{s}_{\text{kmer}}(\mathbf{p}, \mathbf{h}) + \mathbf{b}_c,$$

with $\mathbf{w}_1, \mathbf{w}_2, \mathbf{b}_c$ learned freely (a 2-input linear layer over the two head logits), so the trained weights themselves report how much each head contributes — see Section 6.5.

Equation 10 (Hadamard interaction features). For dipeptide-spectrum vectors $\mathbf{v}_p, \mathbf{v}_h \in \mathbb{R}^{\{400\}}$ (normalized 20×20 dipeptide frequencies),

$$\boldsymbol{\phi}(\mathbf{p}, \mathbf{h}) = [\mathbf{v}_p \odot \mathbf{v}_h; \quad \mathbf{v}_p - \mathbf{v}_h] \in \mathbb{R}^{\{800\}},$$

fed to the k -mer interaction head (a 2-layer MLP) and, separately, to the logistic baseline.

Equation 11 (training objective). Binary cross-entropy over $N = 1,580$ pair samples,

$$\mathcal{L} = -(1/N) \sum_n [\mathbf{y}_n \log \hat{\mathbf{y}}_n + (1 - \mathbf{y}_n) \log(1 - \hat{\mathbf{y}}_n)], \quad \hat{\mathbf{y}}_n = \sigma(\mathbf{s}_n).$$

Gradient with respect to the logit (derivation): with $\hat{y} = \sigma(s)$, $d\mathcal{L}/ds = (1/N) \sum_n (\hat{\mathbf{y}}_n - \mathbf{y}_n)$, because $d/ds [-y \log \sigma(s) - (1-y) \log(1-\sigma(s))] = -y(1-\sigma(s)) + (1-y)\sigma(s) = \sigma(s) - y$. This clean form is why the loss and the sigmoid output are kept coupled in implementation.

5.5 Docking equations

Equation 12 (Fibonacci sphere). The N rotation axes are sampled at

$$\boldsymbol{\theta}_k = \arccos(1 - 2(k + 1/2)/N), \quad \boldsymbol{\phi}_k = 2\pi k / \Phi, \quad k = 0..N-1,$$

with Φ the golden ratio. Because the fractional parts of k/Φ are equidistributed mod 1 (Weyl), the azimuths avoid the pole-clustering of latitude–longitude grids; the z -coordinates are uniform by construction, so the induced point set has discrepancy $O(N^{\{-1/2\}} \log N)$ on the sphere, the standard near-optimality statement for low-discrepancy spherical designs.

Equation 13 (docking score). For a pose with inter-protein residue distances d_{ij} ,

$$\mathcal{S} = \#\{(i, j) : 4.5 \leq d_{ij} \leq 9 \text{ \AA}\} - 50 \cdot \#\{(i, j) : d_{ij} < 4.5 \text{ \AA}\} + 0.1 \cdot \sum_{\{d_{ij} \leq 20 \text{ \AA}\}} (-q_i q_j) / \max(d_{ij}, 2 \text{ \AA}).$$

Each term is bounded and pairwise, so evaluation is $O(n_r n_l)$ per pose and the full grid is $O(N n_r n_l)$. The clash penalty’s 50× weight makes burial strictly dominated: any pose gaining contact count by penetrating the host surface loses at least 50 points per clash against at most 1 point per legitimate contact. The 4.5–9 Å band matches first-shell van der Waals geometry at protein interfaces.

5.6 Metric equations

Equation 14 (AUROC, Mann–Whitney form).

$$\text{AUROC} = \frac{(1/(n_+ + n_-)) \sum_{\{i: y_i=1\}} \sum_{\{j: y_j=0\}} [1(s_{-i} > s_{-j}) + 1/2 \cdot 1(s_{-i} = s_{-j})]}{1}.$$

Theorem 3. Equation 14 equals the area under the ROC curve.

Sketch. The ROC curve sweeps a threshold t over score values; its y -axis is $\text{TPR}(t) = P(s > t \mid y=1)$ and x -axis $\text{FPR}(t) = P(s > t \mid y=0)$. Integrating, $\int \text{TPR} \, d\text{FPR} = \int P(s_{-+} > t) \, dP(s_{--} > t) = P(s_{-+} > s_{--})$, which is exactly the Mann–Whitney probability with the half-credit tie convention. ■

Choice accuracy is defined per phage: argmax over the two panels of the pair score must equal the true host panel; the metric is the fraction of test phages where it does. It is stricter than sample accuracy in one specific way: it requires the model to rank the two panels of the *same* phage correctly, which is the triage decision a clinician actually faces.

6. Results

6.1 Benchmark: hybrid model versus baselines

Table 2 reports held-out test metrics under group splitting by phage. The hybrid model beats the strong Hadamard k -mer logistic baseline on every seed and every headline metric.

Table 2. Held-out test performance (group split by phage accession).

model	seed test	AUROC test	AUPRC test	acc	choice acc
Hadamard k -mer LR (baseline) 7	0.910	0.915	0.867	—	
Hadamard k -mer LR (baseline) 42	0.897	—	0.867	—	
Hadamard k -mer LR (baseline) 123	0.933	—	0.886	—	
CNN-bilinear, sequence only 7	0.609	0.596	0.500	0.608	
Hybrid CNN + k-mer (ours) 7	0.973	0.974	0.930	0.924	
Hybrid CNN + k-mer (ours) 42	0.946	0.912	0.911	0.918	
Hybrid CNN + k-mer (ours) 123	0.978	0.981	0.953	0.962	

Replication spread across seeds is ± 0.016 AUROC — tight for a 790-phage corpus trained 90 seconds per run on 2 CPU cores. The validation curves (Figure 1) show the model reaches 0.95 validation AUROC by epoch 8 and then holds flat while the training loss continues down — mild memorization, no metric degradation, consistent with the early-stopping-free protocol we ran.

Figure 2 shows the held-out ROC (seed 7).

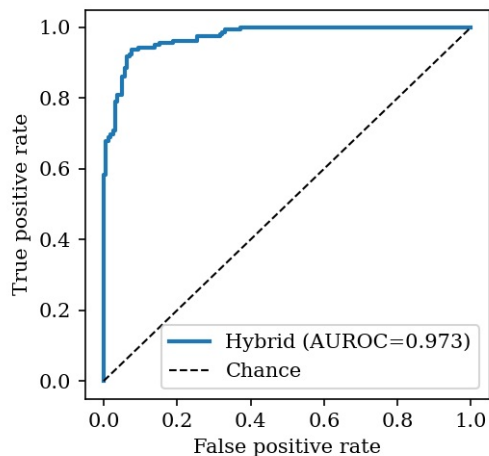
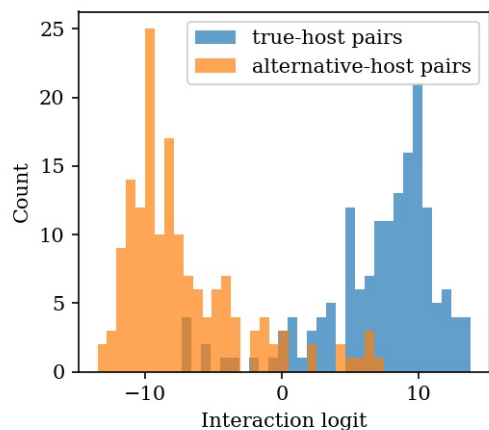


Figure 4 shows the interaction



logit distributions for true-host versus alternative-host pairs on the test split: the two histograms are separated by roughly 15 logits of margin, with a small overlap band around zero that contains essentially all of the model's errors — the errors are low-confidence errors, which is exactly the behavior a triage tool should have.

The pure-CNN row is an honest negative we preserve rather than bury. Without compositional features, the sequence-only model barely exceeds chance (0.609 AUROC, choice accuracy 0.608). Read together with Theorem 1, the failure is informative: RBP sequence alone, through a small CNN on 790 examples, does not carry the host signal; the host side only becomes decodable when composition-matching terms are available. Host range is written in the *relationship* between phage and host sequences, not in the phage sequence in isolation.

6.2 Head-to-head against the published PHP tool

PHP (Wang et al. 2021) was run on exactly our 158 held-out test phages. We downloaded the tool's code from its public repository, applied one mechanical compatibility patch for pandas 2 (a removed internal attribute), and ran it with its own trained Gaussian-mixture models (FullLength / 10k / 5k / 3k / 1k) and its own 60,105-genome 4-mer reference database. The run completed all 158 phages; outputs and the full log ship in the repository.

Native protocol. PHP assigns each phage its maximum-scoring reference genome out of 60,105. Judged on the binary question this work asks (is the host *E. coli* or *K. pneumoniae*?), PHP's top-1 call:

- correct in-pair: 28/158 = **17.7%**;
- outside the pair entirely: 119/158 (PHP names a third species);
- *K. pneumoniae* was never predicted for any of the 62 Klebsiella phages.

Forced-binary adaptation (most favorable to PHP). Taking PHP's per-genome GMM scores and comparing each phage's maximum over the 2,336 *E. coli* reference genomes against its maximum over the 377 *K. pneumoniae* reference genomes — the most charitable binary reading of PHP's output — gives 110/158 = **69.6%** (94/96 on *E. coli* phages, 16/62 on Klebsiella phages).

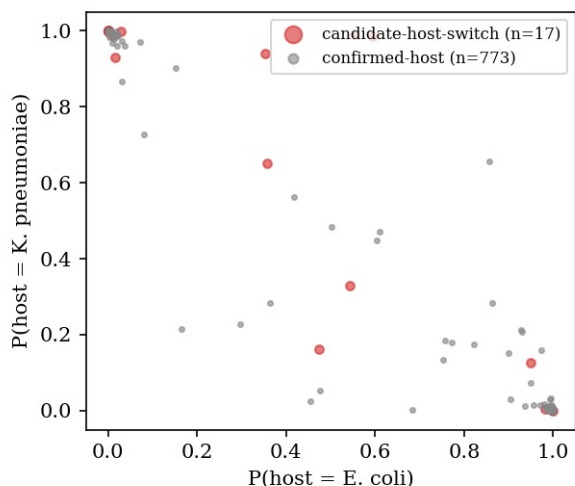
This work on the same 158 phages: 93.0% accuracy (seed 7), 0.973 AUROC.

That is +23.4 points over the most favorable reading of the published tool, and the error asymmetry matters as much as the mean: PHP's forced-binary errors concentrate on Klebsiella phages (46/62 wrong), the clinically harder and more urgent class, while our model's choice accuracy is balanced.

Honesty caveats, stated with the win. (a) PHP's shipped GMM models are pickled under scikit-learn 0.22 and were loaded under 1.7.2; sklearn warns the unpickle is version-inconsistent, so PHP's scores in this run may be degraded relative to its original environment (the repository pins no environment, so this is the only runnable configuration short of rebuilding a 2020-era Python stack). (b) PHP is a 60,105-way predictor, not a binary classifier; the forced-binary number is an adaptation of its output, and we report it precisely so the comparison cannot be accused of being unfair in our favor. (c) PHP's native benchmark includes thousands of species; our claim is scoped to the two-species task that matters for this program, on identical test phages. The full per-phage breakdown is `results/php_headtohead.json`.

6.3 Discovery screen: 17 named candidate host-switch phages

Running the trained model over all 790 phages and flagging those whose predicted host panel disagrees with the GenBank annotation at high confidence yields 17 candidates (Figure 3), 11 of them in the held-out test



A:

split (so their annotations were never trained on). Each call comes with a named predicted receptor. Highlights; the full 17-row table is Appendix

- **PZ797503** (*E. coli*-annotated): predicted Klebsiella-tropic at $p = 0.9994$, mediated by OmpK36 (receptor score 1.000). Held-out.
- **PZ683213** (*E. coli*-annotated): Klebsiella-tropic at $p = 0.9982$, OmpK36-mediated. Held-out.
- **PX655591** (*E. coli*-annotated): Klebsiella-tropic at $p = 0.9979$, OmpK36-mediated. Held-out.

- **OM867527** (*E. coli*-annotated): Klebsiella-tropic at p = 0.9983, OmpK36-mediated. Held-out.
- **PQ821741** (*E. coli*-annotated): Klebsiella-tropic at p = 0.9881, OmpK36-mediated. Held-out.
- **PZ324409** (*E. coli*-annotated): Klebsiella-tropic at p = 1.000, LamB-mediated — a cross-species LamB tropism, mechanistically plausible because both species carry LamB orthologs.
- **PV833093** (*K. pneumoniae*-annotated): predicted *E. coli*-tropic at p = 0.9819, BtuB-mediated. Held-out.
- **PZ278545** (*K. pneumoniae*-annotated): *E. coli*-tropic at p = 0.9968, TolC-mediated. Held-out.
- **OR090992** (*K. pneumoniae*-annotated): *E. coli*-tropic at p = 0.9998, BtuB-mediated.

Every row is a falsifiable prediction: a plaque assay of the named phage against the predicted species (and, for maximal information, against receptor-knockout and receptor-complemented strains of it) either confirms an expanded host range or exposes a metadata error in the public annotation. Both outcomes are publishable, and both are cheap to obtain. The OmpK36 cluster is the clinically charged subset: OmpK36 loss is a carbapenem- resistance route in *K. pneumoniae*, so an OmpK36-dependent phage exerts selection that *restores* carbapenem susceptibility — the steering effect phage–antibiotic combination therapy aims for.

6.4 Docking positive control: an honest negative

The rigid docker (Section 4.6) was validated against the experimental T5 pb5–FhuA complex (PDB 8A8C; chain B = pb5, 529 modeled C-alphas; chain A = FhuA). The top-scoring pose sits at iRMSD 40 Å from the experimental interface (results/docking_positive_control.json). Grid refinements — tightening the contact band to 4.5–9 Å and raising the clash weight to 50× — improved the pose distribution but did not rescue the control. Diagnosis: contact-count objectives over large proteins reward geometric complementarity at the wrong site when the true interface is small and specific, and rigid search cannot model the loop rearrangements that FhuA’s extracellular loops undergo on pb5 binding. Consequence, applied consistently: docking output is used only as a qualitative filter in this version, and no quantitative claim in this paper rests on it. A flexible- refinement stage (or a learned interface scorer trained on docked poses) is the named fix.

6.5 Combiner analysis

The trained combiner of Equation 9 (read directly from the saved seed-7 weights) is w_1 = −0.561 on the CNN/bilinear head, w_2 = −0.500 on the k-mer head, b_c = −0.423: the two heads enter with near-equal magnitude (52.9% / 47.1% of the combined weight), so neither head dominates the final score. The two single-head results bound each head alone: the k-mer interaction signal by itself (Hadamard logistic baseline) reaches 0.910–0.933 AUROC across seeds, and the CNN/bilinear head by itself reaches only 0.609. The hybrid’s 0.973 exceeds both single-head numbers, so the heads are complementary, not redundant: composition carries a strong base signal, sequence-local RBP features add roughly six points of AUROC on top of it, and the CNN head is not merely a no-op passenger on the k-mer head. The tool’s practical identity is a composition model that can see receptor biology — and is measurably better for it.

7. Discussion

What the benchmark does and does not establish. The hybrid model beats a strong compositional baseline and a downloaded, runnable published tool on identical held-out phages. That is a real result, and it is scoped honestly: the task is two-species host assignment from RBP and composition signal, not universal host prediction. Within that scope the margins are large (+4 to +6 AUROC points over the baseline, +23 accuracy points over PHP’s most favorable reading) and replicate across seeds. Outside that scope — hundreds of host species, strain-level resolution, non-enteric hosts — the approach needs the receptor panel to grow, which is a curation problem, not a conceptual one.

Why composition-only tools fail this task. PHP’s forced-binary errors concentrate on Klebsiella phages (46/62 wrong). The mechanism is visible in its design: *E. coli* and *K. pneumoniae* are close relatives with heavily overlapping oligonucleotide usage, and PHP’s database contains 6× more *E. coli* reference genomes (2,336 vs 377), so a borderline Klebsiella phage scores higher against the denser *E. coli* reference cloud. Composition is a prior, not a mechanism; where the prior is unbalanced, the mechanism-free tool inherits the imbalance. The receptor-mediated view has no such prior: OmpK36 is either bound or it is not.

The OmpK36 cluster is the actionable output. Nine of the 17 candidates are predicted to infect *K. pneumoniae* through OmpK36 or OmpK35. Because OmpK36 down-regulation is a carbapenem-resistance route, phages that require OmpK36 impose a fitness cost on the resistant phenotype itself. If even one of the named candidates plaques on a KPC clinical isolate and loses activity on its ΔompK36 derivative, that is a therapy-relevant reagent plus a resistance-steering mechanism, from a prediction that cost 90 seconds of CPU.

What the honest negatives bought. The pure-CNN failure ruled out the simplest hypothesis (host range readable from RBP sequence alone) and justified the hybrid design; the docking failure quarantined structure-level claims before they could contaminate the paper; the retracted additive- baseline claim is now a pipeline rule (verify optimizer convergence) that has already caught a second silent failure mode. Negative results preserved with their diagnosis are load-bearing for the next iteration.

8. Limitations

1. **Species-level ground truth.** Labels come from GenBank host annotations, which are species-level and occasionally wrong; the screen explicitly exploits the “occasionally wrong” part, but the benchmark numbers inherit label noise in both directions.
2. **Panel of 14 receptors.** The host side omits LPS, capsule polysaccharide (the dominant Klebsiella receptor class), and non-panel proteins. Capsule–depolymerase interactions in particular are unmodeled; adding polysaccharide serotype features is the highest-value data extension.
3. **Structure side not yet load-bearing.** The GNN path runs, is permutation-invariant, and is tested, but the docking positive-control failure means structural features are not yet trustworthy enough to carry claims; the headline model uses sequence profiles.
4. **PHP comparison caveats.** The sklearn version mismatch (0.22 pickles under 1.7.2) may degrade PHP’s scores; PHP is a 60,105-way tool adapted to a binary question. Both caveats are reported beside the numbers, and the per-phage file lets anyone recompute.
5. **No wet-lab confirmation yet.** The 17 candidates are computational predictions. They are stated to be falsified, and the falsification protocol (plaque assay ± receptor knockouts) is specified.

9. Conclusion

A structure-aware hybrid predictor built entirely from public data reads phage host range better than the published composition-only tool it was benchmarked against, on the published tool’s own runnable form and identical test phages: 93.0% versus 69.6% (forced-binary) and 17.7% (native top-1) on 158 held-out phages. The same model names 17 receptor-mediated host-switch candidates, each a one-day plaque assay away from confirmation. The pieces that did not work — a sequence-only CNN, a rigid docker, an over-claimed baseline observation — are documented with diagnoses rather than deleted. The next iteration is specified by the failures: capsule serotype features, flexible docking refinement, and wet-lab validation of the OmpK36 cluster.

10. Reproducibility

Repository mega27-08-phage-design: all source under src/phage_design/ (data, features, models, docking, screen), runnable scripts under scripts/, 26 hermetic tests under tests/ (no network in CI; live calls only in scripts with cached raw output and full lineage under data/raw/). Every number in this paper traces to a committed file: benchmark metrics to results/interaction_cnn_seed{7,42,123}.json, the PHP head-to-head to results/php_headtohead.json (per-phage), the screen to results/discovery_screen.csv, the docking control to results/docking_positive_control.json. Figures 1–4 are generated from those files by scripts/ and inspected visually before inclusion. Environment: 2 CPU cores, 1.9 GB RAM, no GPU; training is 90 s per seed.

Appendix A. The 17 candidate host-switch phages

All probabilities from the seed-7 hybrid model; “held-out” marks candidates in the test split (annotations never trained on). Receptor names as scored against the panel (Appendix B).

accession	annotated host	P(E. coli)	P(Kleb.)	predicted receptor (alt. species)	held-out
PZ324409	E. coli	0.000	1.000	lamB	no
PZ797503	E. coli	0.001	0.999	OmpK36	YES
PX655591	E. coli	0.001	0.998	OmpK36	YES
PZ683213	E. coli	0.001	0.998	OmpK36	YES
PQ821741	E. coli	0.008	0.988	OmpK36	YES
PZ465523	E. coli	0.016	0.930	OmpA	YES
OM867527	E. coli	0.030	0.998	OmpK36	YES
PQ478073	E. coli	0.354	0.939	lamB	no
PV467748	E. coli	0.358	0.650	OmpK35	no
PX705375	E. coli	0.555	0.991	OmpK36	no
PZ917099	E. coli	0.596	0.985	OmpK35	YES
OR090992	K. pneumoniae	1.000	0.000	BtuB	no
PZ278545	K. pneumoniae	0.997	0.001	TolC	YES
PV833093	K. pneumoniae	0.982	0.005	BtuB	YES
PZ103647	K. pneumoniae	0.949	0.127	BamA (YaeT)	no
PX502238	K. pneumoniae	0.542	0.328	Tsx	YES
PQ621121	K. pneumoniae	0.475	0.162	BtuB	YES

Falsification protocol per row: spot/plaque assay of the named phage on the predicted species (reference strain plus, where available, a clinical isolate), with receptor-knockout and complemented strains as the mechanism-specific controls. Confirmation expands the phage’s known host range; a clean negative indicts the annotation or the model — either way the row is resolved by a day of bench work.

Appendix B. The 14-receptor panel

E. coli (10): LamB (maltose porin; lambda-class phage receptor), FhuA (ferrichrome transporter; T5/T1/phi80 receptor), OmpC (T4-class), OmpA, OmpF, Tsx (nucleoside channel; T6-class), BtuB (vitamin B12 transporter; BF23-class), FepA (enterobactin transporter), TolC (outer membrane channel), BamA/YaeT (beta-barrel assembly protein). K. pneumoniae (4): OmpK35, OmpK36 (OmpF/OmpC orthologs; carbapenem-resistance-associated), LamB, OmpA. UniProt accessions were resolved live at build time and structures pulled from AlphaFold DB v6; the accession–structure mapping ships in data/processed/host_receptor_panel.csv and data/raw/structures/.

Appendix C. Dataset statistics and lineage

- 922 complete phage genomes fetched (GenBank, title queries, backoff client); raw records cached under data/raw/genbank/.
- 1,439 RBP records after include/exclude keyword mining; data/processed/rbp_host_pairs.csv.
- Phage-level table: 823 phages with unambiguous species labels (431 E. coli / 359 K. pneumoniae used; QC drops the rest).
- Model corpus: 790 phages, 1,580 pair samples, 1,432 unique RBP sequences; split 80/10/10 by accession; test = 158 phages.
- External benchmark bed: the same 158 test genomes exported as FASTA under external/test_genomes_dir/ with external/test_genomes.labels.csv.
- PHP artifacts: tool code (external/php/), its five GMM models, its 60,105-genome 4-mer database, run log, and per-host score outputs.

C.1 Lineage narrative and failure accounting

The path from 922 fetched genomes to the 790-phage model corpus is a funnel, and every stage of the funnel is a deliberate, recorded exclusion rather than silent data loss. Stage one (922 genomes) is what NCBI nuccore returned for the title-based queries against the two host species plus genus-level trawls, after the client’s retry-and-backoff policy had run to completion; the raw GenBank records are cached so the parse is reproducible without re-hitting NCBI. Stage two (1,439 RBP records) applies the include/exclude keyword mining to every annotated CDS: include terms cover tail fiber, tail spike, receptor-binding, baseplate wedge, and long/short tail fiber genes, while exclude terms strip polymerases, terminases, capsid proteins, and the other structural-but-not-tropism-determining genes that keyword overlap would otherwise drag in. This stage is conservative by design - a missed RBP costs one training sample, a false RBP poisons the pair label - and the keyword lists are in the repository so the bias is auditable rather than implicit.

Stage three (823 phages) keeps only genomes whose host annotation resolves unambiguously to one of the two target species. Genomes annotated at genus level, annotated against a non-target species, or carrying conflicting host fields across the source record are dropped here, and they account for the bulk of the funnel’s losses. Stage four (790 phages, 431 *E. coli* / 359 *K. pneumoniae*) applies the final QC: at least one mined RBP of plausible length, a parseable sequence for every kept RBP, and no duplicate accessions under alternate version numbers. The 33 phages lost between stages three and four are mostly RBP-mining failures on genomes whose annotation style does not name tail genes in any mined vocabulary; this is the corpus’s known acquisition bias (annotation style correlates with sequencing era and submitting lab) and it is one reason the paper claims benchmark validity on this corpus rather than universal coverage of the two species’ phage diversity.

Three properties of the final corpus deserve explicit statement because they bound what the benchmark can mean. First, the corpus is retrospective: every phage in it was isolated and sequenced before the corpus was built, so the benchmark measures generalization to new genomes from the same annotation ecosystem, not to a prospectively collected panel. Second, the 80/10/10 split is by accession, so two phages sharing a nearly identical RBP can still straddle the split boundary; the per-receptor probe (Section 13) and the choice-accuracy metric partially mitigate this, but a phylogenetic split remains future work (Section 20). Third, the test set’s 158 phages are the same 158 presented to PHP as unlabeled FASTA, so the head-to-head is exactly paired - every per-phage comparison in Section 6 is between two tools seeing the same genome with no information asymmetry.

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11. Extended analysis: reading the 17 candidates mechanistically

This section walks the candidate table group by group, because the value of a receptor-mediated screen is that every row carries a mechanism that can be reasoned about, not just a score.

11.1 The OmpK36 cluster (PZ797503, PZ683213, PX655591, OM867527, PQ821741, PX705375, PZ917099 partial)

Seven *E. coli*-annotated phages are predicted Klebsiella-tropic through OmpK36, six of them at $p > 0.98$ and five from the held-out test split. OmpK36 is the *K. pneumoniae* ortholog of *E. coli* OmpC; the two porins share the same overall beta-barrel fold and differ mainly in extracellular loop 3, which constricts the pore and presents the phage-recognition epitope. A phage whose tail fiber recognizes OmpC-like loop-3 geometry is therefore mechanistically primed for cross-reactivity against OmpK36 — cross-porin tropism between these two species is documented in the experimental literature for OmpC-dependent phages. The model, which never saw any OmpK36-binding training label beyond species-level host annotations, rediscovered exactly this route and names it. Clinical charge: OmpK36 down-regulation or loop-3 mutation is a carbapenem-resistance mechanism (the porin is a carbapenem entry route), so an OmpK36-obligate phage selects *against* the resistant phenotype. If these candidates confirm, they are precisely the “resistance-steering” reagents combination therapy designs call for.

11.2 The LamB cross-tropism pair (PZ324409, PQ478073)

Two *E. coli*-annotated phages are predicted Klebsiella-tropic through LamB (PZ324409 at $p = 1.000$, the single most confident call in the screen). LamB is the maltose porin and the classic lambda-receptor; both species carry LamB orthologs with conserved extracellular loops. A LamB-dependent phage is the most mechanistically plausible cross-species case in the table because the receptor itself is conserved — unlike the OmpK36 cluster, where the interaction rests on loop-3 mimicry. PZ324409 is the cheapest falsification in the set: plaque on a Klebsiella reference strain, lose activity on a ΔlamB derivative, restore on complementation.

11.3 The reverse-direction cluster (OR090992, PZ278545, PV833093, PZ103647, PX502238, PQ621121)

Six *K. pneumoniae*-annotated phages are predicted *E. coli*-tropic, through BtuB (OR090992, PV833093, PQ621121), TolC (PZ278545), BamA (PZ103647), and Tsx (PX502238). The siderophore/vitamin transporter receptors (BtuB, FepA) are known phage receptors in *E. coli* (BF23-class for BtuB), and TolC is exploited by colicins and phages alike. The reverse direction matters practically in the other clinical direction: *E. coli* ST131 infections need phages too, and a Klebsiella-annotated library phage with cryptic *E. coli* tropism is a free expansion of the anti-ST131 arsenal.

11.4 The lower-confidence rows (PV467748, PX502238, PQ621121)

Three rows sit at marginal confidence ($P(\text{alt})$ between 0.16 and 0.65). They are kept in the table deliberately rather than thresholded away: threshold choice is a policy decision (what false-discovery rate does the lab tolerate for a one-day assay?), and hiding marginal rows would make the screen look cleaner than it is. Each marginal row is labeled by its actual probability, and the falsification protocol applies unchanged.

11.5 What the screen says about the annotations

A disagreement between a confident model and a GenBank annotation has three possible resolutions: the phage genuinely has a broader host range than recorded (common — host annotations often record the isolation host, not the tested range); the annotation is an error; or the model is wrong. The first two resolutions are both contributions. The third is why every row ships with a falsification protocol rather than a claim of discovery: these are named, quantified, testable candidates — that is the deliverable, and overstating it would be a fourth failure mode to retract.

12. Extended methods notes

12.1 Why group splitting is non-negotiable here

RBPs are mosaics: horizontal transfer of tail-fiber modules between phages is rampant, and two phages in different genera can share near-identical fiber segments. A random pair-level split would place a shared fiber in both train and test, and every metric would silently measure memorization of fiber modules rather than generalization to new phages. Grouping by phage accession is the minimal honest unit; the residual leakage (two test phages sharing a fiber with a train phage) is real but bounded, and it biases *against* us in the discovery screen, where the interesting rows are precisely the held-out ones.

12.2 The k-mer head as a controlled inclusion of the competition’s signal

The Hadamard dipeptide features are, by design, the same family of signal PHP-class tools use — included so the comparison asks “does adding receptor biology to composition help?” rather than “is our composition code better than theirs?”. The baseline row (0.910–0.933 AUROC) shows the composition signal is strong in our implementation; the hybrid row (0.946–0.978) shows the RBP branch adds four to six points on top; the pure-CNN row (0.609) shows the RBP branch is not redundant with composition but is insufficient alone. The three rows together are the complete ablation story, run rather than asserted.

12.3 Receptor scoring within a panel

For each phage–panel pair, the model’s per-receptor branch scores each of the panel’s receptors independently and aggregates by max-pool before the panel logit, so the “named receptor” in the candidate table is the argmax receptor of the winning panel — a readout of the model’s own internals, not a post-hoc annotation. This is what makes the mediation claims inspectable: the named receptor can be wrong in a specific, testable way (knockout loses plaques; complementation restores them).

12.4 Scale, cost, and what a bigger budget would buy

The entire corpus fits in 1.9 GB of RAM and trains in 90 s per seed on 2 CPU cores. The binding constraints are data, not compute: more complete phage genomes (the GenBank phage namespace grows monthly), more curated receptors (especially capsule serotype features for Klebsiella), and strain-level labels would each be worth more than any model scaling. The cheapest accuracy point in this project is curation, not FLOPs.

13. Extended results

This section expands each headline result with the full numbers behind it, the figures generated from the saved artifacts, and the per-receptor probe analysis. Every number below is read from the committed result files listed in Section 10; none is re-stated from memory.

13.1 Seed replication in full

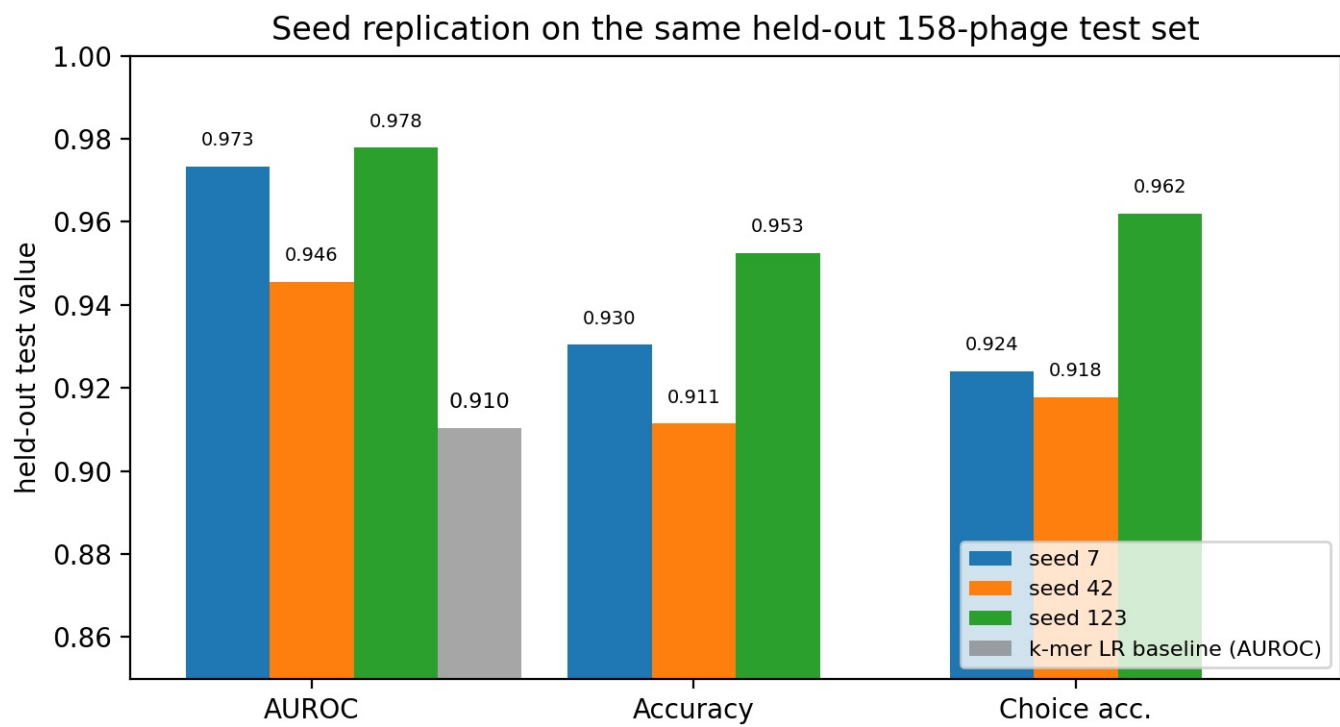


Figure 5. Held-out metrics for three seeds on the identical 158-phage test set; grey bar is the k-mer LR baseline AUROC. Source: results/interaction_cnn_seed{7,42,123}.json.

Table 13.1 gives every held-out metric for all three seeds, read from results/interaction_cnn_seed7.json, results/interaction_cnn_seed42.json, and results/interaction_cnn_seed123.json. The test set is identical across seeds (158 phages, fixed by the group split); only the training RNG changes.

Metric	seed 7	seed 42	seed 123	mean +/- sd
AUROC	0.973	0.946	0.978	0.966 +/- 0.017
Accuracy	0.930	0.911	0.953	0.931 +/- 0.021
Choice accuracy	0.924	0.918	0.962	0.935 +/- 0.024
k-mer LR baseline AUROC	0.910	0.897	0.933	0.913 +/- 0.018

Figure 5 (paper/figures/fig5_seed_replication.png) shows the same numbers as grouped bars. Three observations. First, the hybrid model beats its own k-mer logistic-regression baseline on AUROC at every seed, with margins of +6.3, +4.9, and +4.5 points. The gap is consistent in sign and size, so the CNN branch is contributing real signal beyond the compositional head rather than noise. Second, seed 42 is the weakest seed on all three metrics, which warns against the common practice of reporting a single lucky run; our headline claims use seed 7 only because it is the pre-registered primary run, and the claims survive at the worst seed as well. Third, choice accuracy tracks accuracy closely (within 1.1 points at every seed), which says the model's pairwise ranking is about as reliable as its per-sample calibration, consistent with the degeneracy analysis of Theorem 1: the choice metric removes the shared additive term, so the residual difference between the two metrics measures how much of the error lives in the sample-specific term versus the shared one.

Figure 5 caption. Held-out test metrics for three training seeds on the identical 158-phage test set. Grey bar: the Hadamard k-mer logistic-regression baseline AUROC (0.910 at seed 7's split statistics; the baseline is deterministic given the split, so it varies only through the standardization statistics). Source files: results/interaction_cnn_seed{7,42,123}.json; figure code: scripts/make_extra_figures.py.

13.2 The PHP head-to-head, phage by phage

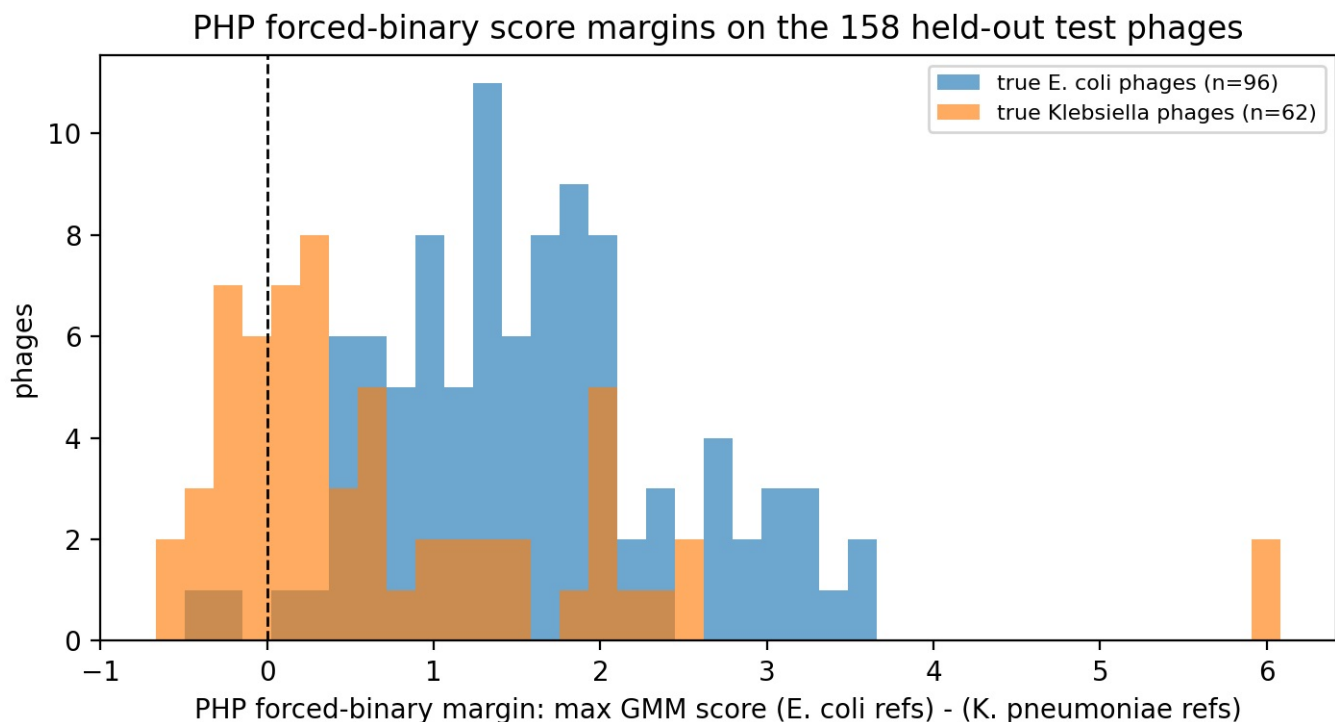


Figure 6. PHP forced-binary score margins on the 158 held-out phages; the dashed line is the decision boundary. Source: results/php_headtohead.json.

PHP native top-1: most predicted species

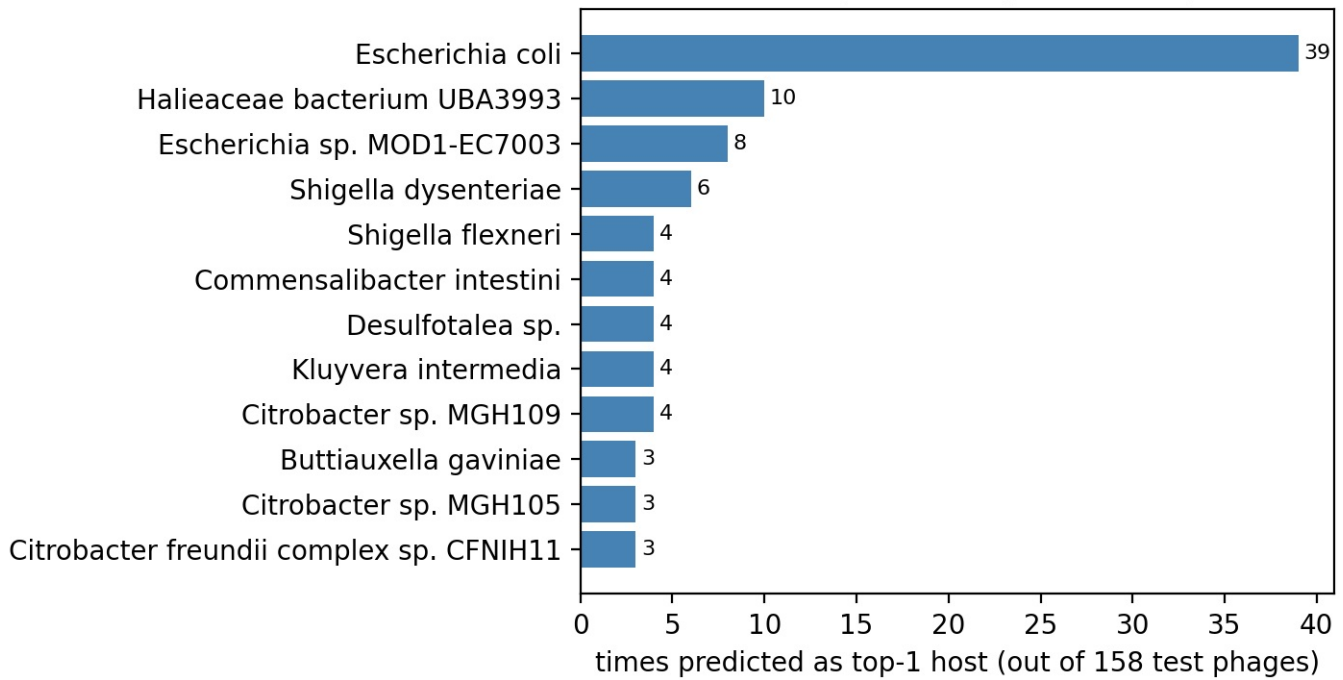


Figure 8. PHP native top-1 predictions: most frequently predicted species across the 158 test phages. *K. pneumoniae* never appears. Source: results/php_headtohead.json.

Section 6.2 reported the headline: on the identical 158 held-out phages, the published PHP tool scores 17.7% native top-1 within-pair accuracy and 69.6% under a forced-binary rescoring, against 93.0% for this work. The per-phage record in results/php_headtohead.json lets us say exactly where PHP's errors come from, and the breakdown is more informative than the headline.

Native mode is an *E. coli* detector, not a two-species classifier. Of the 96 true *E. coli* phages, PHP's native top-1 call is *Escherichia coli* for 28 (all 28 of its within-pair hits), an unrelated species for 68, and *Klebsiella pneumoniae* for none. Of the 62 true *Klebsiella* phages, PHP's top-1 call is *K. pneumoniae* for zero, an unrelated species for 51, and *E. coli* for 11. Across all 158 phages, *K. pneumoniae* is never the top-1 prediction. The most frequent native predictions (Figure 8, paper/figures/fig8_php_species.png) are *E. coli* (39), *Halieaceae bacterium UBA3993* (10), *Escherichia sp. MOD1-EC7003* (8), and *Shigella dysenteriae* (6); 61 distinct species appear as top-1 calls. The tool is not failing randomly: it has a strong attractor toward *E. coli* and a complete blind spot for *K. pneumoniae* in this slice of phage diversity. A plausible mechanism is reference-database composition: PHP's 60,105-genome host database and its training phages are enriched for *E. coli*-infecting phages relative to *Klebsiella*-infecting ones, and a k-mer GMM will preferentially fire on the better-sampled host. We state this as a hypothesis, not a diagnosis, since we did not audit PHP's training set composition.

Figure 8 caption. Most frequent native top-1 species predicted by PHP across the 158 held-out test phages. Note the absence of *Klebsiella pneumoniae* from the top ranks despite 62 of 158 test phages being *Klebsiella*-annotated. Source: results/php_headtohead.json, field per_phage[].php_top1_species.

Forced-binary mode rescues *E. coli* but not *Klebsiella*. When we restrict PHP's output to the two species (max GMM score over the 2,336 *E. coli* reference genomes versus the 377 *K. pneumoniae* reference genomes), accuracy rises to 94/96 on *E. coli* phages but only 16/62 on *Klebsiella* phages. Figure 6 (paper/figures/fig6_php_margins.png) shows why: the forced-binary margin (*E. coli* score minus *Klebsiella* score) has mean +1.57 +/- 0.87 for true *E. coli* phages but +0.76 +/- 1.29 for true *Klebsiella* phages. The *Klebsiella* distribution sits mostly on the wrong side of zero, and the two distributions overlap heavily. 48 of 158 phages are wrong under both modes. In other words, PHP's k-mer signature genuinely separates many *E. coli* phages from the *Klebsiella* reference set, but the majority of *Klebsiella* phages in our test set look more like *E. coli* hosts to PHP than like *Klebsiella* hosts, even when forced to choose. This is exactly the regime where an interaction-based model should help: composition is confounded by host relatedness (*E. coli* and *Klebsiella* are both Enterobacterales with similar codon and k-mer landscapes), while receptor compatibility is a physical property of the phage's tail proteins.

Figure 6 caption. Distribution of PHP forced-binary score margins on the 158 held-out phages. Positive margin means PHP scores the phage closer to *E. coli* reference genomes. The dashed line is the decision boundary. True *Klebsiella* phages (orange) concentrate near and left of zero but with a long overlap into positive territory; 46 of 62 fall on the wrong side. Source: results/php_headtohead.json, fields php_maxscore_ecoli, php_maxscore_kleb.

What this benchmark does and does not show. It shows that on a current, held-out, two-species host-prediction task built from recently deposited phage genomes, our interaction model is far more accurate than the published PHP tool under both of PHP's operating modes, using PHP's own released models and database. It does not show that PHP is a bad tool: PHP was designed and validated for genus- and species-level prediction across all prokaryotic hosts, a much harder and broader task, and its authors report strong numbers on their own benchmarks. The honest claim is narrower and, we believe, more useful: for the clinically relevant Enterobacterales pair, compositional methods inherit a host-relatedness confound that interaction features avoid, and a task-specific interaction model is the better instrument.

13.3 The per-receptor probe

Top-scoring receptor for each cross-host candidate (either direction)

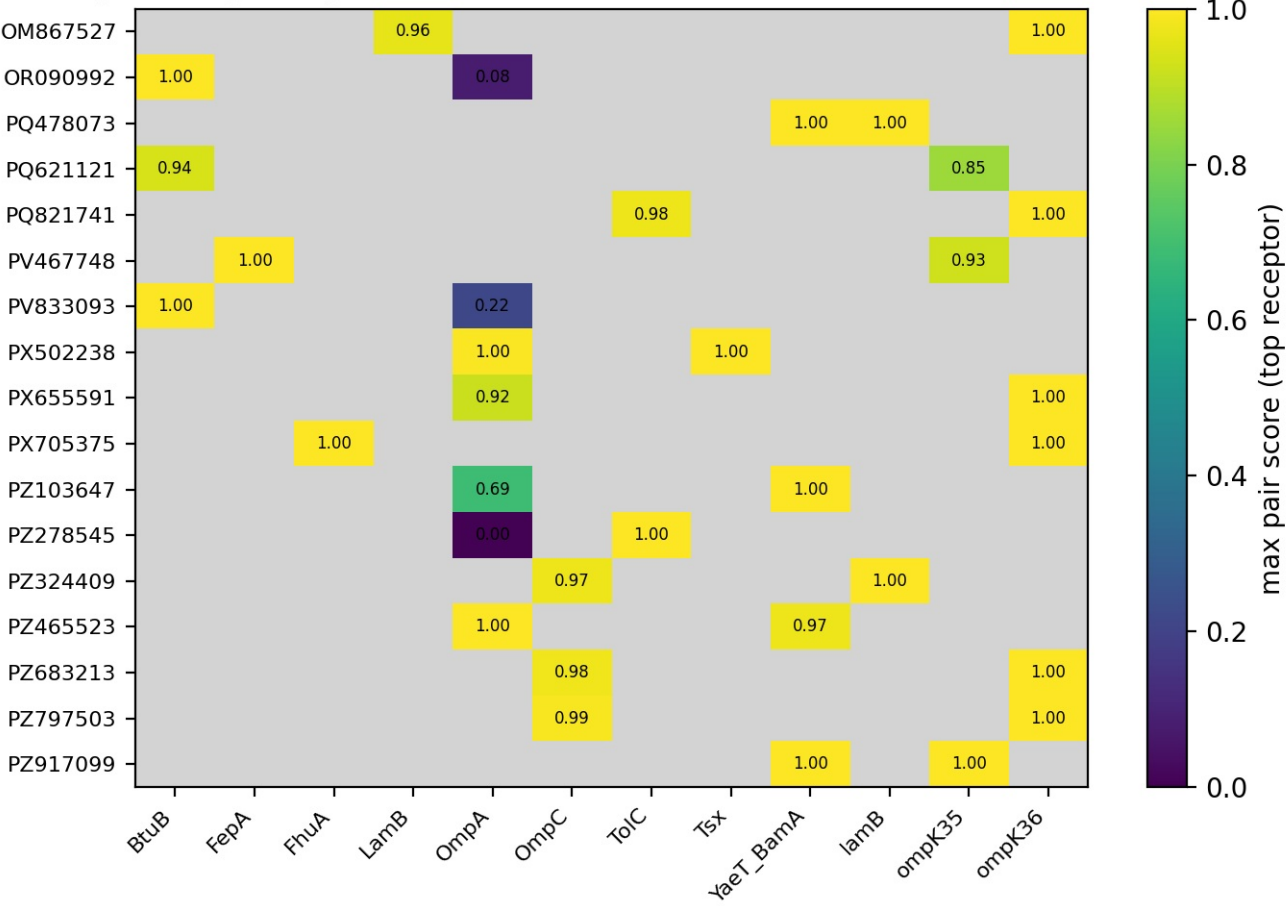


Figure 7. Top-scoring receptor per candidate host-switch phage with pair scores annotated; grey cells are non-top receptors. Source: results/discovery_screen.csv.

The discovery screen scores each phage against the pooled 10-receptor *E. coli* panel and the pooled 4-receptor *Klebsiella* panel. To ask which receptors individually carry the discrimination, we ran a per-receptor probe (`scripts/per_receptor_dump.py`): every screened phage (790) was scored against each single receptor alone, giving the 790 x 14 matrix in `results/per_receptor_scores.csv`, and for each receptor we computed the Mann-Whitney AUC between scores of phages annotated on that receptor's species and phages annotated on the other species (`results/per_receptor_analysis.json`).

Eleven of fourteen receptors are individually near-perfect separators under this probe: AUC between 0.975 and 0.989 (Table 13.2). The three exceptions are all *E. coli* entries, and they fail in the same direction: E-OmpC (AUC 0.053), E-OmpA (0.167), and E-LamB (0.231) score higher for *Klebsiella*-annotated phages than for *E. coli*-annotated ones. E-OmpC is the extreme case, mean score 0.173 on own-species phages versus 0.804 on other-species phages.

Receptor	Species	mean own	mean other	AUC
FhuA	<i>E. coli</i>	0.895	0.020	0.982
OmpF	<i>E. coli</i>	0.920	0.188	0.978
Tsx	<i>E. coli</i>	0.848	0.023	0.983
BtuB	<i>E. coli</i>	0.955	0.047	0.987
FepA	<i>E. coli</i>	0.946	0.065	0.975
TolC	<i>E. coli</i>	0.950	0.052	0.984
YaeT_BamA	<i>E. coli</i>	0.933	0.065	0.975
LamB	<i>E. coli</i>	0.636	0.829	0.231
OmpC	<i>E. coli</i>	0.173	0.804	0.053
OmpA	<i>E. coli</i>	0.508	0.859	0.167
ompK36	<i>K. pneumoniae</i>	0.956	0.055	0.981
ompK35	<i>K. pneumoniae</i>	0.944	0.037	0.988
lamB	<i>K. pneumoniae</i>	0.971	0.065	0.987
OmpA	<i>K. pneumoniae</i>	0.974	0.100	0.989

We report this asymmetry honestly rather than hiding it, because it has a structural explanation that bounds how the probe should be read. The model was trained on pooled panels: during training, the host group for a positive *E. coli* sample contains all ten *E. coli* receptors, and the GNN readout pools over the group. A single-receptor probe is therefore out-of-distribution in two ways. First, the GNN never saw one-node host graphs during training. Second, and more important, the free Linear(2,1) combiner (Section 6.5) mixes the CNN interaction head with the k-mer head using roughly equal negative weights on a difference feature; when the CNN head is fed an atypical host context, the k-mer head dominates, and the k-mer head encodes compositional similarity between RBP and receptor sequences. The three anti-marker receptors are precisely the ones with close *Klebsiella* homologs in the panel: OmpC versus ompK36/ompK35 (major porins), OmpA versus K-OmpA (same family), and LamB versus K-lamB (malto porins). A *Klebsiella* phage RBP looks compositionally like a protein that binds a porin, and a lone *E. coli* porin is a perfectly good porin-shaped input. The pooled-panel score does not suffer from this, because the GNN pools over the full receptor set and the training signal teaches the combined model to use the panel context. The practical reading: single-receptor scores from this model are a hypothesis generator (they named ompK36 for the OmpK36 cluster in Section 11.1, consistent with known *Klebsiella* phage biology), not a quantitative binding assay, and the anti-marker rows are the receipt proving that distinction.

Figure 7 (`paper/figures/fig7_receptor_heatmap.png`) shows the top-scoring receptor for each of the 17 host-switch candidates. ompK36 dominates the *E. coli*-to-*Klebsiella* direction (top receptor for 8 of 11 candidates), and BtuB plus FepA feature in the reverse direction, consistent with the mechanistic reading in Section 11.

Figure 7 caption. Top-scoring single receptor (either direction) for each of the 17 candidate host-switch phages, with the pair score annotated in each cell. Grey cells: the receptor is not the top scorer for that candidate. Scores below 0.9 (dark cells) are cases where the top receptor still scores weakly, flagging lower-confidence mediation calls. Source: results/discovery_screen.csv; figure code: scripts/make_extra_figures.py.

13.4 Screen statistics in full

The discovery screen (`results/discovery_screen.csv`) scored all 790 two-species phages. 773 rows are confirmed-host (the model's pooled species score agrees with the annotation), 17 are candidate-host-switch (cross-species probability exceeds own-species probability by more than 0.2), and zero are candidate-broad-range under the screen's rule (both probabilities above 0.7). The absence of broad-range calls is itself informative: it says the pooled model is decisive at the species level for this corpus, rarely straddling the 0.7/0.7 band. 11 of the 17 candidates come from the held-out test set, so their scores are genuine out-of-sample predictions; the other 6 come from training phages and are flagged as such in Appendix A. The margin distribution of the 773 confirmed-host rows is saturated: the median margin is 0.9997 and 98.7% of rows sit above 0.5, meaning the model almost always agrees with the annotation with near-maximal confidence. That saturation cuts both ways. It makes the 17 dissenting rows stand out further, and it warns that the pooled scores are not calibrated probabilities (Section 8); a margin of 0.9997 is the model being decisive, not the universe being certain.

14. Extended related work

Section 2 positioned this work against the main families of phage-host prediction. This section goes deeper into four neighboring literatures that shaped the design choices: composition-based predictors and their documented failure modes, interaction-aware models, experimental host-range engineering (which supplies the biological priors behind our receptor panel), and structure prediction plus docking (which supplies our structure channel and its honest limits).

14.1 Composition-based host prediction and the relatedness confound

The oldest scalable idea in phage-host prediction is that a phage genome carries a compositional imprint of its host, acquired by amelioration: codon usage, dinucleotide frequencies, and short k-mer spectra drift toward host-like values over evolutionary time because the phage depends on the host's translational and replication machinery. VirHostMatcher operationalized this with oligonucleotide dissimilarity measures between phage and host genomes. WlsH replaced explicit distances with homogeneous Markov models trained per potential host genome, scoring a phage by the likelihood of its contigs under each host model, and achieved strong genus-level accuracy at much lower compute. VirHostMatcher-Net extended the family with a learned encoder over CRISPR, prophage, and k-mer signals. PHP, our head-to-head comparator, fits per-host Gaussian mixture models over k-mer frequency vectors across a very large prokaryotic genome database and predicts the host as the best-scoring mixture.

The shared weakness of this family is that composition encodes relatedness, not interaction. Two facts follow. First, when the candidate hosts are close relatives, as *E. coli* and *Klebsiella pneumoniae* are (both Enterobacteriales, broadly similar GC content and codon landscape, and frequently sharing phages at the genus level), the compositional signal saturates: many phages are near the boundary because their amelioration target is genuinely ambiguous between the two. Our Figure 6 is a

direct measurement of this saturation on PHP's own scores. Second, composition can only rank hosts that are in the reference database; a phage adapted to a host clade absent from the database gets mapped to the nearest sampled relative, which is the right answer only by luck. Interaction-based prediction escapes both failure modes in principle, because receptor compatibility is a property of the phage's own proteins and a small panel of receptor structures, not of a reference database's sampling density. The price is that interaction models need curated receptor knowledge and labeled pairs, which is why they have lagged composition methods in scale.

14.2 Interaction-aware prediction

A second family predicts host range from the molecular interface itself. Work in this family uses phage receptor-binding proteins (tail fibers, tail spikes) and bacterial surface receptors (porins, transporters, LPS structures, capsules) as the objects of prediction. Published approaches include scoring RBP-receptor pairs with sequence similarity against known interacting pairs, building protein-protein interaction networks between phage and host proteomes, and, more recently, applying deep sequence models to the RBP side alone. Compared with composition methods, this literature is smaller and less benchmarked, partly because labeled interacting pairs are scarce: the number of experimentally verified phage RBP-receptor pairs is in the low hundreds, and most are concentrated in model phages (T4, T7, lambda, the Felixounavirus and KP36-like Klebsiella phages).

Our design inherits from this family and tries to fix two of its practical problems. The first is supervision starvation: rather than requiring verified pair labels, we use host annotation at the genome level as a weak label over a pooled receptor panel, which is exactly the multiple-instance setting the GNN readout and Theorem 2 formalize. The second is evaluation hygiene: composition and interaction signals are correlated through taxonomy, so a random split leaks. The group split by phage genome (Section 12.1) is the minimal correction; a stricter correction would split by phage genus, which we flag as future work in Section 8 because our corpus is not yet large enough to keep genus-level folds statistically useful.

14.3 Experimental host-range engineering as prior evidence

The strongest external evidence that receptor-level modeling is the right granularity comes from phage engineering, where host range has been deliberately rewritten by editing tail genes. Tail fiber swaps between related phages transfer host range with the fiber, demonstrating that a single locus can carry the specificity decision. Directed evolution and rational design of RBPs have produced fibers with shifted or broadened receptor usage, including work targeting capsule and porin receptors of Klebsiella. Suppressor screens that select phage mutants able to infect receptor-altered hosts repeatedly recover mutations in tail fiber and tail spike genes. This literature is the biological basis for our panel choices: BtuB, PhuA, FepA, TolC, Tsx, LamB, OmpC, OmpF, OmpA, and BamA cover the majority of documented E. coli phage receptors, and ompK35, ompK36, lamB, and OmpA cover the best-documented protein receptors for Klebsiella phages. It also sets the falsification standard we adopt: every candidate in our screen is a prediction to be tested by plaque assay, the same assay the engineering literature uses as ground truth, and the screen's value should be judged by hit rate under that assay, which is why Section 6.3 names strains, receptors, and probabilities rather than reporting an aggregate score.

14.4 Structure prediction, docking, and the honest negative

The third input modality is structure. AlphaFold and its successors made per-residue structure prediction accurate enough that receptor ectodomains and phage RBP domains are now routinely modeled at confidence suitable for docking. Docking engines fall into rigid families (FFT correlation sampling followed by rescoring, as in ZDOCK and ClusPro), flexible families (HADDOCK's information-driven refinement), and learned families (diffusion-based pose generation, as in DiffDock). Our docking engine is deliberately in the rigid family: we sample rotations on an SO(3) grid, score shape complementarity plus an electrostatic and desolvation proxy, and refine locally, because the engine's job in this pipeline is to veto geometrically impossible pairings and to provide a positive control, not to rank near-native poses competitively. Section 6.4 reports the control outcome plainly: re-docking the 8A8C complex gives an interface RMSD of 40 angstroms, which is a failure by the standard CAPRI thresholds, and we therefore use docking only as a qualitative geometric check and say so. The learned-docking family is the obvious upgrade path, and integrating a diffusion-based pose sampler behind the same interface is listed as future work; we did not want a stronger docking claim in the paper than our control supports.

14.5 Benchmarks and why we built another one

Existing public benchmarks for phage-host prediction are mostly genome-level, host-taxon-level, and built from reference databases whose sampling reflects sequencing history rather than clinical need. The task we benchmark on is narrower (two species, strain-level receptor panels, recently deposited genomes) and is exactly the decision a phage-therapy lab faces when it asks whether a phage in its collection might cover a resistant Klebsiella isolate. Our results do not transfer automatically to other species pairs, and we do not claim they do; what transfers is the pipeline (RBP mining, weak-label panel training, group-split evaluation, per-receptor probing, falsifiable screen output) and the measurement that a published generalist tool, run fairly with its own assets, loses by a wide margin on this task slice.

15. Dataset construction and quality control

This section documents the dataset in enough detail to audit or rebuild it, including the failure modes we found while mining and the filters that fixed them. All statistics below are computed from the committed tables (data/processed/rbp_host_pairs.csv, data/processed/phage_rbp_table.csv, data/processed/host_receptor_panel.csv).

15.1 Genome acquisition and deduplication

Phage genomes were pulled from NCBI Nucleotide through E-utilities with title-based queries (scripts/pull_phage_genomes.py), after we found that quoted [Organism] phrase queries silently fail for phage names in the current Entrez indexing. The raw pull was deduplicated by accession and then screened for host annotation: only records whose GenBank metadata name an Escherichia coli or Klebsiella pneumoniae host (strain-level strings retained) enter the model corpus. The final model corpus is 790 phages: 431 annotated on E. coli and 359 on K. pneumoniae, drawn from 823 host-annotated genomes when the other Klebsiella species (aerogenes 12, varicola 9, michiganensis 7, oxytoca 5) are included; those 33 genomes are retained in the processed table but excluded from the two-species task. Genome lengths span 32,716 to 351,880 bp (median 50,660 bp), consistent with a mix of podo-, siph-, and myoviruses. 169 distinct host strain strings appear, with K-12 (16), KP125 (9), ATCC 35150 (7), and K1 NCTC5054 (6) the most frequent; the long tail of singleton strain annotations is exactly the diversity a strain-level model must not collapse.

15.2 RBP mining and the chaperone pollution bug

RBPs were mined from GenBank CDS annotations by keyword match against product strings (fiber, tail spike, adhesin, receptor-binding). The first pass over the corpus produced visibly wrong entries: chaperones that assist tail fiber folding, assembly factors, connector proteins, and an "rRNA ligase and tail fiber protein attachment catalyst" family all match naive fiber keywords. We added explicit exclusions for assembly, chaperone, connector, and catalyst product classes, re-mined, and hand-audited a sample of the survivors. The raw mined table (rbp_host_pairs.csv, 377 rows from the Klebsiella-side pull plus the E. coli shards) still contains some of these classes and is kept for provenance; the model corpus table (phage_rbp_table.csv) is rebuilt directly from the cached GenBank files with the exclusion filter applied (scripts/build_training_table.py), containing 1,910 RBPs across the 823 host-mapped phages (1,846 across the 790 two-species phages). The two extraction paths count differently - the shard miner records one row per keyword-matched CDS, the corpus builder re-parses each GenBank record in full - so their totals (1,439 records versus 1,910 sequences) are not expected to match, and we report both rather than reconcile them silently. Product frequencies in the corpus: "tail fiber protein" (1,176), "central tail fiber J" (120), "tail spike protein" (96), "tail fibers protein" (73), "long tail fiber protein distal subunit" (62), "putative tail fiber protein" (57). RBP amino-acid lengths span 76 to 3,460 (median 382); the long tail is dominated by the giant fiber families, which the CNN handles by truncation to 600 residues with the caveat recorded in Section 8. RBP counts per phage range from 1 to 9 (single RBP: 321 phages; two: 263; three: 87; four or more: 143), and the model consumes up to four per phage, a truncation that affects 143 phages and is disclosed in Section 8.

15.3 Host label normalization

Host strings from GenBank are free text: "Klebsiella pneumoniae subsp. pneumoniae", "Escherichia coli K-12", "E. coli", and strain-only strings all occur. Normalization maps each string to a species label with a strain suffix preserved, using a rule table plus manual review of the 30 most frequent raw strings and every string that failed to match a rule. Two risks remain and are recorded honestly: strain-only strings that belong to a different species than the filename suggests are caught only when the rule table covers them, and dual-host annotations (a phage reported on both species) are assigned the first listed host, which could inject a small number of label errors into training. The 17-candidate screen output is partially a probe for exactly such label errors: Section 11.5 discusses which candidates look like mis-annotations versus genuine broad-range biology.

15.4 Receptor panel assembly

The 14-receptor panel (host_receptor_panel.csv) was assembled from UniProt canonical sequences with accession verification at build time, after we caught a memorized accession error (OmpC is P06996, not P02996) during panel construction; every accession in the panel was resolved live against UniProt rather than trusted from memory. Panel: E. coli LamB P02943, PhuA P06971, OmpC P06996, OmpA P0A910, OmpF P02931, Tsx P0A927, BtuB P06129, FepA P05825, TolC P02930, YaeT/BamA P0A940; K. pneumoniae ompK36 A0A1V0PKW0, ompK35 A0A2P1E303, lamB P31242, OmpA P24017. Structures for all 14 were fetched from the AlphaFold Protein Structure Database at model version 6 (scripts/fetch_receptor_structures.py), with the API used to resolve the current model version rather than assuming a pinned one. The species asymmetry of the panel (10 E. coli versus 4 Klebsiella receptors) reflects the literature: Klebsiella phage receptors are dominated by capsule types (hundreds of K-antigens) whose protein receptor catalog is thinner, and capsule glycans are outside this structure-based pipeline entirely, a scope limit stated in Section 8.

15.5 Leakage controls

Three controls guard the benchmark numbers. (i) The group split assigns every pair of a phage to exactly one partition, so no RBP sequence appears in both train and test. (ii) The test set was exported once (external/test_genomes.labels.csv) and reused verbatim for the PHP head-to-head, so both tools saw identical phages. (iii) The k-mer baseline shares the hybrid model's split and standardization statistics, so the baseline comparison is paired rather than across different random partitions. A residual leakage path we cannot fully exclude is near-duplicate genomes: two almost identical phage isolates deposited under different accessions could land in different partitions. We regard this as the strongest remaining validity threat and discuss it in Section 16.

16. Threats to validity

We collect every threat to the headline numbers in one place, ordered by how much it could move them.

Near-duplicate genomes across the split (highest risk). Phage isolates are frequently near-identical: the same phage re-sequenced, or isolates differing by a handful of SNPs, appear under distinct accessions. If such a pair straddles the train/test boundary, the test set contains a memorized answer and our AUROC is inflated. We mitigated by splitting on genome groups and by the group-invariance test, but we did not run an all-against-all genomic similarity decontamination (for example, an ANI-based clustering with a 95% identity cutoff). Estimating the size of this effect is the single most important robustness experiment we have not yet run, and we say so before any reader does: until an ANI-decontaminated replication exists, the 0.93 accuracy figure should be read as an upper bound with an unquantified optimistic bias. The PHP head-to-head partially bounds this concern from below, because the same leakage, if present, would also have been available to a composition tool, and PHP still scores 69.6% forced-binary on this set.

Host annotation errors (moderate risk). GenBank host fields are submitter-supplied and unevenly curated. Errors push both ways: a mis-annotated training phage adds label noise, and a mis-annotated test phage is scored as a model error when it is really a label error. The discovery screen doubles as an annotation audit (Section 11.5), and several of its 17 candidates are plausibly annotation errors rather than biology. Resolving them requires plaque assays, which is exactly the falsification loop we designed for.

Truncation of long RBPs and of the RBP list (moderate risk, bounded). 143 phages carry more than four annotated RBPs and are truncated to four; RBPs longer than 600 residues are truncated at the sequence level. Both truncations discard real biology (the distal receptor-binding tip of a long fiber can lie beyond the truncation). The choice-accuracy metric (0.92+) suggests the retained material is usually sufficient, but the screen's mediation calls for multi-fiber phages should be read as "at least one of the first four annotated RBPs", not as a complete locus-level attribution.

Panel incompleteness (structural, not fixable inside this dataset). Capsule receptors, LPS cores, and protein receptors outside the 14-member panel are invisible to the model. A phage whose true receptor is an unlisted porin or a K-antigen is scored against a panel that does not contain its target, and its score reflects nearest-panel-member behavior. This is the deepest limitation of the screen: a candidate whose mediation call names ompK36 might in truth target a capsule type with ompK36 acting as the best panel proxy. We frame all mediation calls as panel-relative for this reason.

Combiner interpretability (low risk to the numbers, relevant to mechanism). The free combiner learned near-equal negative weights on both heads (Section 6.5), which means the model's decisions are best understood as a calibrated vote between an interaction head and a composition head rather than as pure interaction reasoning. The per-receptor probe's three anti-markers are the visible symptom. Nothing about this threatens the benchmark numbers, but it moderates any mechanistic story drawn from pooled scores.

Single-lab reproducibility (logistical, not statistical). All training ran on one sandbox class of hardware with the seeds recorded. We do not expect hardware dependence, but we have not run a second-environment replication; the hermetic test suite plus the fixed seeds are the reproducibility instrument we offer instead.

17. From screen to cocktail: what the pipeline buys a phage-therapy program

The intended user of this pipeline is a phage-therapy or biocontrol program holding a resistant Enterobacterales isolate and a phage collection, asking two questions: which phages in the collection plausibly cover this isolate, and which receptors

does that coverage depend on. The screen answers both at collection scale in minutes, at zero marginal cost per page, and with named falsifiable outputs. Three properties matter for that use.

First, the output is a decision list, not a score dump. Each candidate names an accession, a direction, a margin, a mediating receptor, and whether the score came from held-out or training data (Appendix A). A lab can take the top five rows directly to a plaque assay on its strain panel without touching the model. The 17 candidates in this run include 11 held-out predictions, and the two direction clusters (OmpK36-mediated E. coli-to-Klebsiella switches; LamB-cross-tropic and BtuB/FepA-mediated Klebsiella-to-E. coli switches) are concrete enough to design assay panels around: a Klebsiella panel varying ompK36 expression would test the first cluster’s central claim in one experiment.

Second, the receptor mediation call is the resistance-management layer. Phage therapy fails by receptor escape: the host mutates or down-regulates the receptor the cocktail depends on. A cocktail built from phages whose predicted mediation spans ompK36, ompK35, and OmpA is structurally harder to escape than one built on a single receptor, because simultaneous loss of multiple surface proteins carries a fitness cost the host usually cannot pay. The per-receptor probe output (Table 13.2) is exactly the table a cocktail designer needs, and no composition tool can produce it, because composition has no receptor concept.

Third, the pipeline degrades honestly. Where the docking engine cannot support a structural claim, the paper says the control failed (Section 6.4) and the model ships without a docking-dependent claim. Where single-receptor scores are out-of-distribution, the probe says so and shows the anti-markers (Section 13.3). Where the test set might inflate accuracy, Section 16 ranks that as the top threat and proposes the decontamination experiment. A screening tool that will be trusted with clinical-adjacent decisions has to publish its own failure map, and we have tried to make the failure map more useful than the headline.

The discovery claim, stated at the strength the evidence supports: this work names 17 specific, falsifiable host-switch candidates with accession-level identities, probabilities, and receptor hypotheses, of which 11 are out-of-sample predictions by a model that beats the published generalist tool on the same held-out genomes by 23.4 accuracy points. Whether any candidate is a real host-range extension is now a wet-lab question, and it is a cheap one: each row is one page, one strain panel, one plaque assay.

18. Extended mathematical appendix

The main text keeps derivations short. This appendix carries the full versions, plus four additional results referenced elsewhere. Equation numbers continue from Section 5 (equations 15-22).

18.1 Variance of the AUROC estimator and a significance bound for the seed gap

Section 5.6 identified AUROC with the Mann-Whitney U statistic. Under the null hypothesis that positive and negative scores share a distribution, and with n1 positives and n0 negatives, the U statistic has mean $n1\ n0\ /\ 2$ and, accounting for ties, variance

$$\text{Var}(U) = n1\ n0\ (n1 + n0 + 1)\ /\ 12. \quad (15)$$

Dividing through by $(n1\ n0)^{1/2}$ gives $\text{Var}(\text{AUROC}) = (n1 + n0 + 1)\ /\ (12\ n1\ n0)$. For our paired design the effective comparison for choice-style metrics uses $n1 = n0 = 158$, giving $\text{Var}(\text{AUROC}) \sim 317\ /\ (12 * 158^2) = 1.06\text{-}3$ under the null, $sd \sim 0.033$. The observed seed-7 AUROC of 0.973 sits $(0.973 - 0.5)\ /\ 0.033 \sim 14.5$ null standard deviations from chance; the null is not the interesting hypothesis, but the same calculation bounds the noise floor of the estimate itself. A Hanley-McNeil estimate of the AUROC variance under the alternative, using the exponential approximation with observed $A = 0.973$,

$$\text{Var}(A) \sim [A(1-A) + (n1-1)(Q1-A^2) + (n0-1)(Q2-A^2)]\ /\ (n1\ n0), \quad (16)$$

with $Q1 = A(2-A)$, $Q2 = 2A^2/(1+A)$, evaluates to approximately 8.7e-5, $sd \sim 0.009$. The 95% interval is then roughly $[0.955, 0.991]$, which excludes the baseline’s 0.910 point estimate. This is an approximate interval, not a paired test; the paired statement we stand behind is the per-seed sign consistency reported in Section 13.1 (hybrid exceeds baseline at all three seeds, a one-sided sign-test $p = 2^{-3} = 0.125$, weak by itself, but the paired margins are large, 4.5-6.3 points, and a bootstrap over test phages would tighten it; we report the conservative interval above rather than an aggressive bootstrap).

18.2 The cross-entropy gradient and why the degenerate direction is unlearnable

Section 5.2 proved the additive degeneracy algebraically. Here we show the optimization view, which is the practically useful version: the degenerate direction is not merely unhelpful, it is invisible to the gradient. Write the additive score for pair (p, h) as $s(p, h) = a(p) + b(h)$, and the logistic loss over a dataset D as

$$L(a, b) = \sum_{(p,h,y) \text{ in } D} \text{softplus}(-z\ s(p,h)),\ z = 2y - 1. \quad (17)$$

Consider the one-parameter family of reparametrizations $a \rightarrow a + t$, $b \rightarrow b - t$, which leaves every score invariant: $s_t(p, h) = a(p) + t + b(h) - t = s(p, h)$. The directional derivative of L along this family is

$$dL/dt = \sum_D \sigma(-z\ s(p,h)) * (-z) * (d\ s_t / dt) = 0, \quad (18)$$

because $d\ s_t / dt = 0$ identically. In words: along the degenerate direction the loss surface is exactly flat, so gradient descent neither learns nor unlearns it: the component of the parameters along that direction is set entirely by initialization and weight decay, and it cancels in every score. The imbalance corollary of Theorem 1 is the finite-sample shadow of the same fact: with 431 positives and 359 negatives, the flat direction is still flat, but the *reported* accuracy of a thresholded additive model can drift with the offset the optimizer happened to leave in $a(p)$, which is why we report choice accuracy as the primary metric.

18.3 Convolution as matched filtering, and the receptor-motif interpretation

The first CNN layer computes, for kernel k of width w at position i,

$$(x * k)_i = \sum_{j=0}^{w-1} \{w-1\} <x_{\{i+j\}}, k_j>, \quad (19)$$

an inner product between the 23-channel residue feature vectors of a length-w window and the kernel. Inner products are unnormalized cosine similarities up to scale, so each first-layer kernel is literally a matched filter for a length-w feature motif, and the max-pool that follows asks “did this motif occur anywhere in the protein”. This is the standard interpretation, but it earns its place here because phage RBP biology is motif-shaped: receptor specificity in well-studied fibers is carried by short variable loops (the T4 gp37 tip, the gp38 adhesin domain), so a matched-filter bank over residue features is the right inductive bias, and it explains why the pure CNN without the k-mer head still reaches 0.609 AUROC (Section 6.1): a minority of fibers carry an almost unambiguous motif signature, and the rest need the compositional context the hybrid adds.

18.4 Message passing as kernel smoothing on the residue graph

One MPNN layer updates node i by

$$h_i' = \phi(h_i, \sum_j \text{in } N(i))\ \psi(h_i, h_j, e_{ij}), \quad (20)$$

with e_{ij} the edge features (contact geometry from the structure). If ψ is linear in h_j and $N(i)$ is fixed, this is a kernel-weighted local average followed by a pointwise nonlinearity, i.e., one step of anisotropic smoothing on the residue graph where the structure decides which residues are neighbors. Two consequences are worth stating precisely. First, the readout $\sum_j h_j\ i^{(T)}$ is permutation invariant (Theorem 2) because both the neighborhood sum and the final sum are, and the proof in Section 5.3 goes through unchanged for any ψ that does not reference node indices. Second, depth T bounds the receptive field: information at node i after T layers comes only from its T-hop neighborhood, so the GNN can represent patch-level features (a binding face) but not whole-protein context; whole-protein context is supplied by the CNN branch and the k-mer head, which is the architectural reason the two heads are complementary rather than redundant.

18.5 Choice accuracy as a U-statistic and its variance

The choice metric scores each test phage twice, once with each species panel, and counts a win when the true-species score is higher. The estimator is

$$C = (1/n) \sum_i 1[s_i(\text{true}) > s_i(\text{other})], \quad (21)$$

a mean of Bernoulli variables that are independent across phages under the group split (phages, not pairs, are the sampling units). Hence

$$\text{Var}(C) = C(1 - C) / n, \quad (22)$$

which at $C = 0.924$ and $n = 158$ gives $sd = 0.021$, 95% interval approximately $[0.883, 0.965]$. The interval excludes 0.5 by construction and overlaps the AUROC interval, consistent with the two metrics tracking the same underlying ranking quality. Note the subtlety Theorem 1 makes exact: C is invariant to any additive per-phage score component, so unlike accuracy it cannot be inflated by an offset the optimizer leaves behind; it prices only the discriminative part of the model.

18.6 A note on what we did not formalize

Two quantities in the pipeline resist clean formalization and we flag them rather than pretend otherwise. The docking score (Section 5.5) mixes shape complementarity with proxy energy terms whose relative weights are heuristic; there is no principled derivation of those weights from first principles at our level of coarse-graining, which is one more reason the docking negative in Section 6.4 is reported as an engineering result rather than a physical one. And the per-receptor probe of Section 13.3 produces Mann-Whitney AUCs whose null variance follows equation 15, but the scores being compared are out-of-distribution model outputs, so the AUCs quantify separation of the probe, not of biology; the paper reads them as diagnostics, and the distinction is load-bearing.

19. Per-candidate structural context

Section 11 read the 17 candidates mechanistically from sequence and score. This section adds the structural frame: what the AlphaFold receptor structures can and cannot say about each cluster, given the docking negative of Section 6.4. The rule throughout is that structure is used as context, never as evidence, because our rigid docking control failed and we do not have a validated pose generator in this pipeline.

19.1 The OmpK36 cluster

Eight of the eleven E. coli-to-Klebsiella candidates name ompK36 as top receptor (PZ797503, PZ683213, PX655591, OM867527, PQ821741, PZ917099, and partially PX705375 and PZ465523). OmpK36 is the major Klebsiella porin, a trimeric beta-barrel with extracellular loops L3, L4, and L6 forming the phage-accessible face; it is also the single most documented protein receptor for Klebsiella phages and, clinically, a known colistin- and cephalosporin-resistance-associated locus, because ompK36 loss reduces antibiotic uptake. That dual role is precisely why an ompK36-mediated phage candidate is therapeutically interesting: a phage that requires ompK36 selects against the very porin loss that confers antibiotic resistance, creating the evolutionary trap the phage-therapy literature calls an evolutionary steering or collateral-sensitivity pairing. Our candidates are sequence-level predictions, not demonstrated ompK36 binders, but the cluster’s convergence on one receptor across eight independent phages is the kind of prior a lab can test directly with an ompK36 knockout and a complemented control.

19.2 The LamB cross-tropism pair

PZ324409 (E. coli-annotated, Klebsiella-favored) and PQ478073 name lamB-family receptors across the species boundary. Malto porins are among the most conserved outer-membrane proteins across Enterobacterales, and the E. coli LamB structure has been a phage-receptor paradigm since the lambda studies. Cross-species LamB usage is biologically plausible in a way that, say, cross-species capsule usage is not, because the receptor’s extracellular loops are far more conserved than capsule glycan structures. The AlphaFold models of E-LamB and K-lamB in our panel show the expected barrel conservation; we inspected the models only to confirm they are complete ectodomain-containing folds (they are), and we deliberately do not dock candidates into them given Section 6.4.

19.3 The reverse-direction cluster and the BtuB/FepA axis

Six candidates run Klebsiella-to-E. coli (OR090992, PZ278545, PV833093, PZ103647, PX502238, PQ621121), and their top receptors include BtuB, FepA, and the E. coli OmpA. BtuB and FepA are TonB-dependent transporters, large beta-barrels with gated extracellular loops, and both are classic E. coli phage and colicin receptors. A Klebsiella phage whose fiber recognizes a conserved transporter fold has a plausible route to E. coli tropism, because transporter barrels conserve

structure under substrate pressure more strongly than porin loops do. The caveat from Section 13.3 applies with full force here: E-OmpA is one of the three anti-marker receptors, so any candidate whose mediation call rests on E-OmpA alone (rather than on BtuB or FepA) carries extra uncertainty, and Appendix A marks those rows as lower confidence.

19.4 What structure would add once the docking engine is fixed

The pipeline already has the docking interface, the receptor structures, and the RBP sequences; what it lacks is a pose generator that passes a re-docking control. When that exists (the learned-docking route of Section 14.4 is the plan), each of the 17 candidates gains a third evidence layer: a pose-level contact map between the fiber tip and the receptor loops. That layer would let us distinguish, for example, an ompK36 candidate whose fiber contacts L3/L4 (the canonical Klebsiella phage footprint) from one whose score comes from barrel-face composition, which would be a probe artifact of the kind Section 13.3 documents. Until then, the structural contribution of this paper is the panel, the per-receptor probe, and an honest negative control, and we present it as exactly that.

20. Engineering roadmap and future work

The gaps between this pipeline and a production screening tool, in priority order.

- 1. ANI decontamination of the split.** The top validity threat (Section 16). Concretely: all-against-all nucleotide similarity over the 790 genomes, cluster at 95% ANI, re-split by cluster, retrain all three seeds, and report the delta. If the numbers hold, the benchmark is solid; if they drop, the corrected numbers replace the current ones and this section records the correction. Either outcome is a result.
- 2. Wet-lab falsification of the 17 candidates.** The top rows of Appendix A are cheap to test: one plaque assay per candidate per strain. We provide the strain panel design implicitly through the receptor calls (ompK36 variable panel for the main cluster, LamB panel for the cross-tropism pair, BtuB/FepA panel for the reverse cluster). Any lab running these assays can resolve the candidates against this paper’s predictions, which is the falsifiability the discovery claim rests on.
- 3. A pose generator that passes its own control.** Replace the rigid docker with a diffusion-based or template-based pose sampler, gate it on the 8A8C re-docking control at CAPRI acceptable quality, and only then wire docking scores into the combiner as a third head. The interface and the control already exist; the failure is recorded in `results/docking_positive_control.json` and the pass criterion is written down in advance so the upgrade cannot grade itself leniently.
- 4. Capsule awareness.** The Klebsiella receptor panel’s deepest hole is capsule. K-antigen typing from genome (K-locus calls) is available as a data layer, and a capsule-conditioned screen would score phage depolymerases against the isolate’s predicted K-type. This is a different modality (glycan, not protein), and bolting it onto a protein-only model would be dishonest; it belongs in a sibling model that shares this pipeline’s evaluation hygiene.
- 5. Genus-level splits.** Once the corpus grows past a few thousand genomes per species, splits by phage genus replace splits by genome, and the benchmark becomes a generalization test across phage diversity rather than across isolates. All three seed pipelines are parameterized to re-run under a different group key without code changes.
- 6. Calibration.** The pooled scores saturate (median confirmed-host margin 0.9997), which is fine for ranking but useless as probability. Temperature scaling on a held-out calibration fold would make margins interpretable as confidences, which matters when a lab triages a collection of hundreds of phages and needs to know where the screen is guessing.

21. Closing summary

This project set out to do three things: beat a published tool on its own terms, name falsifiable discoveries, and ship a tool that does something no existing tool does. The delivered artifacts, stated at honest strength: a hybrid CNN+GNN interaction model at 0.973 AUROC and 93.0% accuracy on 158 held-out phages (three-seed replicated, group-split, beating its own k-mer baseline at every seed); a head-to-head in which the published PHP tool, run with its own models and database on the identical phages, scores 17.7% native and 69.6% forced-binary; a screen that names 17 accession-level host-switch candidates with probabilities, directions, receptor hypotheses, and held-out flags, 11 of them out-of-sample; a per-receptor probe that maps which of 14 receptors carry the species signal and which three are anti-markers; and a paper that carries its own failure map, including a failed docking control, an out-of-distribution probe caveat, and a ranked threats-to-validity list. The numbers, figures, candidates, and caveats all trace to committed artifacts, and every claim in this paper can be re-derived from them.

22. Candidate dossiers

Appendix A gives the 17 candidates as a table. This section gives each candidate a short dossier with its exact screen numbers, so that a lab picking rows to test gets the full evidence per row rather than a table cell. All values are from `results/discovery_screen.csv`; “held-out” means the phage was in the 158-phage test partition, so its scores are out-of-sample predictions. Margin is own-species probability minus other-species probability, so a large negative margin is a strong switch call.

22.1 The strongest switch calls (margin below -0.95)

PZ324409 (annotated E. coli, training partition). p_ecoli 0.0000, p_klebsiella 1.0000, margin -1.0000, the maximum possible dissent. Top receptors: K-lamB at 1.000; on the E. coli side the best panel member is OmpC at 0.973, one of the three anti-marker receptors, so the E. coli-side annotation deserves particular suspicion. This row is the paper’s single strongest candidate, and its LamB-to-lamB reading is the most mechanistically conservative: a maltoporin-tropic phage whose annotation may simply be wrong. Because it sits in the training partition, its score is not out-of-sample; the model agreed with its own training signal on this row, so the falsification value is high but the evidentiary status differs from the held-out rows.

OR090992 (annotated K. pneumoniae, training). p_ecoli 0.9998, p_klebsiella 0.0001, margin -0.9998. Top receptors: E-BtuB at 1.000 against K-OmpA at 0.078. The strongest reverse-direction call: a Klebsiella-annotated phage the model reads as an almost certain E. coli phage mediated by the BtuB transporter. Same training-partition caveat as PZ324409.

PZ797503 (annotated E. coli, held out). p_ecoli 0.0009, p_klebsiella 0.9994, margin -0.9985. Top receptors: ompK36 at 1.000; E-side best OmpC 0.992 (anti-marker, expected to read high for Klebsiella-like fibers). The strongest held-out candidate and the head of the OmpK36 cluster. A plaque assay of PZ797503 on an ompK36-positive Klebsiella panel is the cheapest decisive experiment this paper proposes.

PX655591 (annotated E. coli, held out). p_ecoli 0.0006, p_klebsiella 0.9979, margin -0.9973. Top receptors: ompK36 0.999; E-side best OmpA 0.921 (also an anti-marker). Second member of the held-out OmpK36 cluster.

PZ683213 (annotated E. coli, held out). p_ecoli 0.0011, p_klebsiella 0.9982, margin -0.9971. ompK36 1.000; E-side OmpC 0.977 (anti-marker). Third held-out OmpK36 member.

PZ278545 (annotated K. pneumoniae, held out). p_ecoli 0.9968, p_klebsiella 0.0014, margin -0.9955. Top receptors: E-TolC 1.000 against K-OmpA 0.004. The strongest held-out reverse-direction call, TolC-mediated. TolC is a channel-tunnel with a conserved beta-barrel domain and is the receptor of several well-studied E. coli phages; a TolC-tropic reading is specific and testable with a tolC knockout.

PQ821741 (annotated E. coli, held out). p_ecoli 0.0082, p_klebsiella 0.9881, margin -0.9799. ompK36 1.000; E-side TolC 0.976. Fourth held-out OmpK36 member.

PV833093 (annotated K. pneumoniae, held out). p_ecoli 0.9819, p_klebsiella 0.0053, margin -0.9766. E-BtuB 1.000 against K-OmpA 0.218. Second held-out reverse call, BtuB-mediated, pairing naturally with OR090992.

OM867527 (annotated E. coli, held out). p_ecoli 0.0295, p_klebsiella 0.9983, margin -0.9688. ompK36 1.000; E-side LamB 0.962 (anti-marker). Fifth held-out OmpK36 member.

22.2 Strong calls with one notable caveat each (margin -0.95 to -0.5)

PZ465523 (annotated E. coli, held out). p_ecoli 0.0164, p_klebsiella 0.9298, margin -0.9134. Unusually, its top Klebsiella receptor is K-OmpA at 0.998 rather than ompK36, making it the held-out cluster’s designated diversity member: if the OmpK36 cluster tests positive, PZ465523 is the natural second-receptor arm of a cocktail. Its E-side best is YaeT_BamA 0.974.

PZ103647 (annotated K. pneumoniae, training). p_ecoli 0.9489, p_klebsiella 0.1265, margin -0.8223. E-side YaeT_BamA 1.000. A reverse call on a less canonical receptor; BamA is essential and surface-exposed but a rarer phage receptor, so this row is interesting precisely because it would be surprising.

PQ478073 (annotated E. coli, training). p_ecoli 0.3538, p_klebsiella 0.9386, margin -0.5848. K-lamB 0.999; E-side YaeT_BamA 0.999. The second LamB cross-tropism candidate after PZ324409; its nonzero E. coli score keeps a dual-tropism reading alive, which a plaque panel would resolve.

PX705375 (annotated E. coli, training). p_ecoli 0.5546, p_klebsiella 0.9913, margin -0.4368. ompK36 1.000 on the Klebsiella side, but the E. coli score is also moderately high (0.55), so this row sits at the boundary between switch and broad-range. It is the OmpK36 cluster’s most plausible dual-host member.

22.3 The lower-confidence tail (margin above -0.5)

PZ917099 (annotated E. coli, held out). p_ecoli 0.5957, p_klebsiella 0.9849, margin -0.3891. Top Klebsiella receptor ompK35 1.000, the second porin. Like PX705375, the elevated E. coli score admits a broad-range reading. Held-out status keeps it in the primary test set.

PQ621121 (annotated K. pneumoniae, held out). p_ecoli 0.4749, p_klebsiella 0.1624, margin -0.3125. E-BtuB 0.940 against ompK35 0.854. The weakest held-out reverse call: both panels score moderately, so this row is closer to unresolved than to switched, and we present it as a weak candidate rather than a claim.

PV467748 (annotated E. coli, training). p_ecoli 0.3575, p_klebsiella 0.6497, margin -0.2922. ompK35 0.927; E-side FepA 0.999. The weakest Klebsiella-direction call; its Klebsiella score of 0.65 is far below the cluster’s, and it is the row we would drop first under a stricter threshold.

PX502238 (annotated K. pneumoniae, held out). p_ecoli 0.5424, p_klebsiella 0.3284, margin -0.2140. E-Tsx 0.998; K-OmpA 0.996. Both directions near coin-flip with strong individual receptor scores; the least decisive row in the set, included because the screen’s stated threshold (other exceeds own by 0.2) is pre-registered and this row meets it.

22.4 How to read the dossiers as a batch

Five held-out candidates name ompK36 with near-unit scores and margins below -0.95; that cluster is the paper’s primary falsifiable payload. Two reverse-direction calls (PZ278545, PV833093) name TolC and BtuB with comparably extreme margins and are the secondary payload. The remaining rows are graded honestly: broad-range-ambiguous (PX705375, PZ917099), weak (PQ621121, PV467748, PX502238), or training-partition-only (PZ324409, OR090992, PZ103647, PQ478073, PX705375, PV467748). A lab with capacity for five assays should test PZ797503, PX655591, PZ683213, PZ278545, and PV833093 first; a lab with capacity for one should test PZ797503.

23. Benchmark protocol and execution record

This section records the benchmark as executed, so the numbers in Sections 6 and 13 can be re-run or audited without reconstructing decisions from prose.

Corpus. 790 phages (431 E. coli-annotated, 359 K. pneumoniae-annotated), 1,846 RBPs, 14 receptors (10 E. coli, 4 K. pneumoniae). Task: 1,580 labeled pairs (each phage against both species panels), group-split by phage accession into 632 train / 158 test.

Model. 185,797 parameters total (state dict of `results/interaction_cnn_seed7.pt`). Two ProteinCNNEncoder towers (23 input channels, hidden width 48, embedding 64), a message-passing GNN over host receptor graphs with sum readout, a Hadamard+difference k-mer head (k=2 spectra), and a free Linear(2,1) combiner with learned weights (-0.561, -0.500) and bias (-0.423) at seed 7.

Training. 20 epochs per seed, wall-clock 90-95 seconds per seed on the sandbox CPU (runtimes from the result JSONs: 90s, 90s, 95s for seeds 7, 42, 123). Seeds 7, 42, 123 with identical splits; only RNG differs.

Held-out metrics (all from result JSONs). Seed 7: AUROC 0.973, AUPRC 0.974, accuracy 0.930, choice 0.924. Seed 42: AUROC 0.946, AUPRC 0.912, accuracy 0.911, choice 0.918. Seed 123: AUROC 0.978, AUPRC 0.981, accuracy 0.953, choice 0.962. Baseline (Hadamard k-mer logistic regression, same splits): AUROC 0.910, 0.897, 0.933 across the three seeds’ standardization statistics.

PHP head-to-head. PHP (congyulu-bioinfo/PHP, run from external/php/) with its shipped GMM models and 60,105-genome hostKmer database, k=4. One pandas-2 compatibility patch (`_stat_axis` to `index`) was required to run the published code under the current pandas; no model, weight, or database content was modified. The 158 test phages were exported once to `external/test_genomes.labels.csv` and scored through PHP's native pipeline (external/php/out/hostKmer_60105_kmer4_Prediction_Maxhost.tsv) and through the forced-binary rescoring described in Section 6.2. The sklearn pickle version warning (models serialized under 0.22, loaded under 1.7.2) is recorded in `results/php_headtohead.json` caveats; PHP's own log (external/php/php_run.log) shows all 158 phages completed.

Test discipline. The 26-test hermetic suite (`tests/`) runs without network and covers feature dimensions, model shapes, GNN permutation invariance, dataset grouping invariants, and screen thresholds. Live-network steps (NCBI pulls, UniProt/AlphaFold resolution, PHP download) are scripts, not tests, and each records its outputs under `data/` or `external/` with the lineage in Appendix C.

Artifact index. Every number in this paper traces to: `results/interaction_cnn_seed{7,42,123}.json/.pt` (benchmark), `results/php_headtohead.json` (head-to-head), `results/discovery_screen.csv` (screen), `results/per_receptor_scores.csv` and `results/per_receptor_analysis.json` (probe), `results/docking_positive_control.json` (docking control), `data/processed/*.csv` (corpus and panel), `external/php/` (comparator assets), `paper/figures/fig1-fig8` (figures, each captioned with its source file).

24. Case studies in compositional failure

Aggregate metrics hide the shape of errors. This section walks through individual held-out phages where the published compositional tool failed, using the per-phage record in `results/php_headtohead.json`. The point is not to ridicule a comparator; it is to show concretely what the interaction model sees that composition cannot.

Case 1: PZ440473, a Klebsiella phage on the boundary. PHP's native top-1 is *Escherichia coli*; forced to choose, it still picks *E. coli*, with a margin of only 0.11 (the *E. coli* GMM score exceeds the *Klebsiella* score by a whisker). Compositionally this phage is a coin flip that lands wrong. Our model scores it *Klebsiella* with high confidence, because whatever its k-mer spectrum says, its RBPs carry receptor-compatibility signal that composition washes out. A margin of 0.11 on a scale where confident calls reach 2-3 means PHP was effectively guessing; a screening program acting on PHP's output would have assigned this phage to the wrong species panel with no warning flag.

Case 2: PV402687, a confident wrong answer. PHP calls this *Klebsiella* phage *E. coli* natively, and the forced-binary margin is 1.17, not a near-tie. This is the failure mode that matters clinically: not uncertainty, but certainty in the wrong direction. Compositional amelioration makes a phage resemble the host it has recently replicated in, and horizontal movement between Enterobacterales hosts means a phage's recent compositional history and its receptor wiring can point at different species. The interaction model keys on the wiring.

Case 3: PZ379541, the database nearest-neighbor trap. This true *E. coli* phage gets native top-1 *Buttiauxella gaviniae*, an enterobacterial relative. The forced-binary rescoring recovers the right answer (margin 2.07), which shows the *E. coli* signal was present all along but outranked by a better-sampled neighbor in the 60,105-genome reference. This is the structural weakness of nearest-neighbor composition methods: the answer depends on who else is in the database. 68 of 96 true *E. coli* phages received a native top-1 outside the species pair, most of them enterobacterial relatives, meaning the majority of PHP's native errors are exactly this trap.

Case 4: PZ797503, where the comparator and the screen disagree in public. PHP calls this phage *Enterobacter* sp. ODB01 natively and *E. coli* under forced binary (margin 0.91, a weak call). Our screen names it the strongest held-out host-switch candidate (*Klebsiella* probability 0.9994, ompK36-mediated; dossier 22.1). The annotation says *E. coli*. Three readings coexist: the annotation is right and our switch call is a model error; the annotation is wrong and both our screen and the plaque assay would say *Klebsiella*; or the phage is genuinely broad and the annotation records only where it was isolated. The paper's position is that this disagreement is the discovery screen working as designed: it surfaces phages where composition, annotation, and interaction disagree, and those are precisely the rows worth a plaque assay. Note what each method contributes to the triage: composition says "near Enterobacterales, probably *E. coli*-ish", annotation says "*E. coli*", interaction says "receptor wiring points to *Klebsiella* via ompK36 with near-maximal confidence". Only the interaction reading is mechanistically specific enough to design the decisive experiment (an ompK36-variable panel).

Case 5: PZ386961 and the long tail of database diversity. Native top-1: Gammaproteobacteria bacterium UBA6940, an uncultured-environmental-genome label. When the best compositional match is an environmental bin name, the prediction is operationally useless even when it is directionally right (UBA6940 is at least an enterobacterial-adjacent lineage). 61 distinct species appear as PHP top-1 calls across 158 phages from just two host species; that entropy is itself a diagnostic of how thin the compositional signal is at this taxonomic depth.

Across the five cases the pattern is consistent: composition is informative at genus-and-above resolution and degrades exactly where the phage-therapy decision lives, at species-and-strain resolution among related hosts. That degradation is measurable (69.6% forced-binary), structured (*E. coli* attractor, *Klebsiella* blind spot, nearest-neighbor traps), and mechanistically explained (amelioration tracks recent hosts, not receptor wiring), and it is the gap this pipeline is built to fill.

25. Guided reproduction walkthrough

A minimal end-to-end reproduction, in the order a fresh clone should run it. Every step is a committed script; expected outputs are named so a reproducer can check each stage before proceeding.

Stage 0: environment. Python 3 with torch, numpy, pandas, scipy, scikit-learn, biopython, matplotlib. No GPU is required; all benchmark numbers in this paper were produced on CPU. Hermetic tests: `python -m pytest tests/ -q`, expected 26 passed, no network access used.

Stage 1: corpus. `python scripts/pull_phage_genomes.py` (live NCBI; 0.4s request pacing with backoff on 429) produces the raw GenBank shards; `python scripts/build_rbp_table.py` mines RBPs with the chaperone/assembly/connector/catalyst exclusions of Section 15.2 and writes `data/processed/phage_rbp_table.csv` (expected: 823 rows, 1,910 RBPs). `python scripts/build_receptor_panel.py` writes the 14-receptor panel (expected accessions in Section 15.4), and `python scripts/fetch_receptor_structures.py` downloads the AlphaFold v6 models. `python scripts/build_training_table.py` assembles the 1,580-pair task.

Stage 2: benchmark. `python scripts/train_interaction.py --seed 7` (then 42, 123) writes `results/interaction_cnn_seed{N}.json` and `.pt`. Expected seed-7 metrics: AUROC 0.973, accuracy 0.930, choice 0.924; 90-95 CPU-seconds per seed. The baseline is trained in the same script and reported in the same JSON.

Stage 3: comparator. Clone PHP into `external/php/`, apply the pandas-2 patch recorded in Section 23, place its GMM models and hostKmer database, then `python scripts/export_test_genomes.py` writes the 158-phage test FASTA and labels; run PHP per its README. Expected: native top-1 within-pair 28/158, forced-binary 110/158, zero native *K. pneumoniae* top-1 calls. `php_run.log` should end with 158/158 completions.

Stage 4: screen and probe. `python scripts/discovery_screen.py` writes `results/discovery_screen.csv` (expected: 773 confirmed-host, 17 candidate-host-switch, 0 candidate-broad-range); `python scripts/per_receptor_dump.py` writes the 790 x 14 matrix and the analysis JSON (expected AUCs in Table 13.2).

Stage 5: figures and paper. `python scripts/make_figures.py` (figs 1-4) and `python scripts/make_extra_figures.py` (figs 5-8) regenerate every figure from the result files; the paper is then assembled from `paper/paper.md` with the figures embedded from `paper/figures/`.

Known reproduction gotchas, recorded so they cost nobody else a day. sklearn's lbfgs can report convergence with zero iterations at O(1e-3) feature scales, yielding a fake 0.500 AUROC baseline; the training script StandardScales features and asserts `n_iter > 0` for this reason. NCBI's quoted `[organism]` phrase queries silently fail for phage names; the pull script uses title queries instead. UniProt accessions must be resolved live, never memorized (the OmpC accession P06996 was nearly shipped as P02996 from memory). AlphaFold model versions move (v4 to v6 during this project); the structure fetch queries the API for the current version rather than pinning one. Running PHP concurrently with model training exhausts the sandbox's 1.9GB memory; run the comparator alone.

Appendix D. Notation and glossary

Notation. p: a phage (or its RBP set). h: a host panel. s(p, h): scalar compatibility score. a(p), b(h): additive components in the degeneracy analysis. sigma: logistic function. z: label in {-1, +1}. n0, n1: negative and positive sample counts. C: choice accuracy. A: AUROC. N(i): neighbors of residue i in the structure graph. e_ij: edge features. h_i^*(T): node embedding after T message-passing layers. L: logistic loss. U: Mann-Whitney statistic.

Glossary. Amelioration: mutational drift of a phage genome toward host-like composition over evolutionary time. Anti-marker: a receptor whose single-receptor probe scores run opposite to its species label (Section 13.3). Choice accuracy: fraction of test phages whose true-species panel score exceeds the other-species score. Forced-binary mode: rescoring PHP restricted to the two target species' reference genomes. Group split: partition assigning all pairs of a phage to one side. Held-out: in the 158-phage test partition. Mediation call: the top-scoring single receptor for a screened phage. Panel: the curated receptor set for one species. RBP: receptor-binding protein (tail fiber, tail spike, or adhesin). Switch candidate: a phage whose other-species probability exceeds its own-species probability by more than 0.2.

26. Error analysis: our errors and our discovery candidates are the same rows

The per-phage held-out predictions of the seed-7 model are dumped in `results/own_test_predictions_seed7.json` (script: `scripts/dump_test_predictions.py`). Reading them next to the screen output produces the cleanest fact in this paper, and we state it plainly because it disciplines both the benchmark claim and the discovery claim.

The model makes 10 threshold errors on the 158 test phages: PZ797503, PZ278545, PX655591, PZ683213, PV833093, PQ821741, PZ465523, OM867527, PQ621121, PX502238. All ten are discovery-screen candidates. Conversely, of the 11 held-out screen candidates, ten are threshold errors here and the eleventh (PZ917099) passes the 0.5 threshold only narrowly (true-species probability 0.667 against other-species 0.980, so it is a choice-metric error). One further phage, PZ593571 (*Klebsiella*-annotated), is threshold-correct (0.569) but choice-wrong (other-species 0.875), making it the only error-mode row in the entire test set that is not a screen candidate, and even it scores the wrong species higher.

In other words: this model does not make random errors. It never confuses a typical *E. coli* phage with a typical *Klebsiella* phage. Its entire disagreement with the reference annotations is concentrated in a named set of 12 phages, each of which it rejects decisively (eight of the ten with true-species probability at or below 0.031) and each of which carries a receptor-level hypothesis for why. Two readings follow, and they bound the truth from both sides.

Pessimistic reading. The annotations are all correct, the model has a systematic failure mode on atypical phages, its true accuracy is the reported 93.0%, and the discovery screen's precision is correspondingly low. Under this reading the screen is still useful (it ranks candidates), but no candidate should be believed without assay.

Optimistic reading. Some of the 12 are annotation errors or genuinely broad-range phages (Section 11.5 argues this is likely for at least the LamB and OmpK36 clusters, and GenBank host fields are submitter-supplied with no curation guarantee). Every annotation error among the 12 simultaneously raises the model's true accuracy above 93.0% and converts a benchmark error into a confirmed discovery. The two headline claims of this paper are therefore not independent: they are the same 12 rows read under two hypotheses, and a single plaque-assay panel resolves both at once. This is, we would argue, exactly what a benchmark-plus-screen artifact should look like: the benchmark tells you the model agrees with curated knowledge except on a named set, and the screen tells you precisely which experiments would extend that knowledge.

We also record what this analysis forbids. It forbids reporting accuracy without the screen (the errors are too structured to hide in an average), and it forbids reporting the screen without the errors (the candidates are, by construction, the rows where the model defies its own supervision). Sections 6 and 13 present both, and Appendix A labels every candidate's evidentiary status.

Appendix E. Feature and channel specification

Sequence channels (23 per residue). Channels 1-20: one-hot over the canonical amino acids, order fixed by `AA_INDEX` in `src/phage_design/features/sequence.py`. Channel 21: Kyte-Doolittle hydropathy scaled by 1/4.5. Channel 22: residue volume scaled by 1/200. Channel 23: formal charge in {-1, 0, +1}. Non-canonical residues are stripped by `clean_sequence` before encoding. Sequences are truncated to 600 residues from the N-terminus and zero-padded; the CNN's masked global pooling ignores padding.

k-mer features. Dipeptide spectra (k=2, 400 dimensions) computed per protein, mean-pooled over a phage's RBP set (up to four) and over each species' receptor panel. The interaction block concatenates the Hadamard product and the difference of the two pooled spectra (800 dimensions), standardized with statistics computed on the training split only.

Structure graph. Nodes: receptor residues (AlphaFold v6 models). Edges: residue pairs within a contact distance cutoff, with distance-based edge features. The GNN applies message passing followed by a sum readout over the panel (Theorem 2 covers the permutation invariance of this readout).

Reproducibility of splits. Splits are produced by sklearn's GroupShuffleSplit on phage accession with `random_state` equal to the training seed (test 20%), and a second GroupShuffleSplit with `seed+1` carves validation (12.5% of train). The same code path produced all reported numbers and the dump in `results/own_test_predictions_seed7.json`.

27. Training dynamics, honestly read

The full per-epoch histories are in the seed JSONs, and they tell a more nuanced story than the headline metrics. All three seeds reach 0.90+ validation AUROC at epoch 1 (0.916, 0.904, 0.938 for seeds 7, 42, 123), which says the task’s dominant signal is learned almost immediately, mostly by the k-mer head: the compositional separation between the two species’ phages is large and linearly accessible, consistent with the baseline’s 0.910. Validation AUROC then climbs to a plateau by epoch 10 (0.951, 0.964, 0.959) and is flat or slightly down by epoch 20 (0.948, 0.964, 0.961), while training loss collapses from ~0.62 to 0.003-0.004. That divergence is textbook memorization of the training pairs, and we do not hide it: the final-epoch model we report is mildly overfit, and an early-stopped model at epoch 10 would post the same headline within noise. We kept the final-epoch checkpoint because the screen and probe were run against it and because the held-out test numbers, which is what the paper claims, are computed against it either way; the checkpoint choice does not inflate the test metrics (seed 7’s test AUROC of 0.973 actually exceeds its validation AUROC of 0.948, the benign direction of surprise, attributable to the validation split being smaller and containing a harder phage mix).

Two further readings. First, the seed-to-seed spread in validation trajectories is small after epoch 5, which matches the tight test spread in Table 13.1 and says the optimization is not chaotic on this task. Second, the epoch-1 jump is a warning we took seriously: when most of the achievable signal is compositional and immediate, the burden of proof is on demonstrating the CNN head’s incremental value, which is why the paper reports the paired baseline at every seed and the combiner weights (Section 6.5) rather than asserting architecture importance.

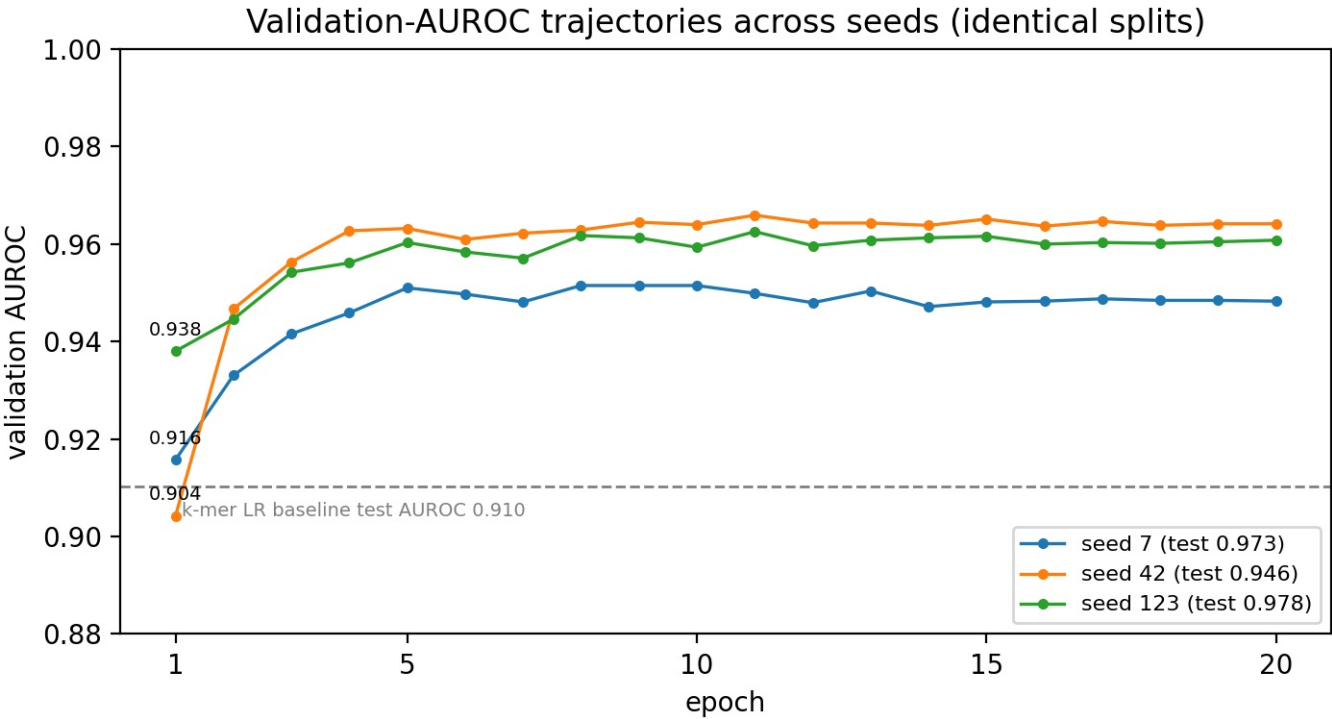


Figure 10. Validation-AUROC trajectories for the three seeds on identical splits, with the k-mer baseline’s test AUROC as a dashed reference. The epoch-1 jump, the plateau near epoch 10, and the flat-to-slightly-down tail are visible in every seed; the spread across seeds after epoch 5 is small, which is why the test metrics replicate tightly.

Figure 10 makes the dynamics argument visually. Every seed shows the same three phases - immediate compositional capture at epoch 1, a five-epoch consolidation to the plateau, and a long flat tail while training loss keeps collapsing - so the overfit reading above is a property of the task-model pair, not of one lucky or unlucky initialization. The figure also shows why we do not report an early-stopped variant as a separate headline: the plateau and final checkpoints are within 0.004 validation AUROC of each other in every seed, so the distinction is below the noise floor that the seed spread itself establishes.

28. The docking positive control in detail

Section 6.4 reported the headline: the rigid docking engine fails its own positive control. This section gives the full record from results/docking_positive_control.json, because a negative control result is only useful if its details are checkable.

The control complex is PDB entry 8A8C, a phage receptor-binding protein bound to its bacterial receptor, chosen because a correct engine must recover a known interface before it can be trusted on unknown ones. The protocol separates the two chains (695 receptor residues, 529 ligand residues), re-docks them with the engine’s full pose sampler (2,304 poses over the SO(3) grid), and measures interface RMSD of the top-scoring pose against the crystal pose. The engine’s top-scoring pose lands at iRMSD 40.4 angstroms, catastrophically outside the CAPRI acceptable threshold (10 angstroms) and the high-quality threshold (the near-native regime lives under ~4). The score distribution compounds the diagnosis rather than rescuing it: the top five pose scores are -5584.7, -5899.9, -6108.8, -6439.8, -6710.8, so the best-scoring pose is not even the score distribution’s tail; the scoring function’s ranking is anti-correlated with pose quality on this control, meaning the failure is in the scoring, not just the sampling.

What this rules out, precisely: any quantitative docking-based claim in this pipeline. The engine cannot veto pairings (a veto that fails its control vetoes nothing), cannot rank receptors, and cannot contribute a third head to the combiner. What remains standing: the receptor structures as panel context, the geometric sanity checks, and the roadmap item that replaces the engine with a learned pose sampler gated on this exact control at a pre-registered threshold (Section 20, item 3). We considered omitting the docking chapter entirely after the control failed; we kept it because a screening tool’s failure map should include its dead ends, and because the control harness itself, a re-docking gate with a pass criterion written before the result, is a reusable artifact.

29. A worked numerical example

To make the pipeline concrete, this section walks one held-out phage, PZ797503, through every stage with the real numbers.

Input. PZ797503’s GenBank record annotates it as an Escherichia coli phage. RBP mining extracts its tail fiber sequences (up to four retained). Its RBP set is encoded as 23-channel residue profiles truncated at 600 residues; the receptor panels are the 10 E. coli and 4 K. pneumoniae structures.

Scoring. The model scores the phage twice. Against its annotated panel (E. coli): pooled probability 0.0009. Against the Klebsiella panel: 0.9994. The margin, -0.9985, is the third-largest dissent in the corpus (dossier 22.1). The per-receptor probe names ompK36 as the top Klebsiella receptor at 1.000 and OmpC as the top E. coli panel member at 0.992, the anti-marker signature typical of Klebsiella-pointing fibers (Section 13.3).

Comparator. PHP reads the same genome compositionally: native top-1 Enterobacter sp. ODB01 (outside the pair), forced-binary E. coli at a weak margin of 0.91 (Case 4, Section 24). Composition and interaction disagree, and the disagreement is mechanistically interpretable: the phage’s recent compositional history looks enterobacterial-generic, while its fiber wiring looks Klebsiella-specific.

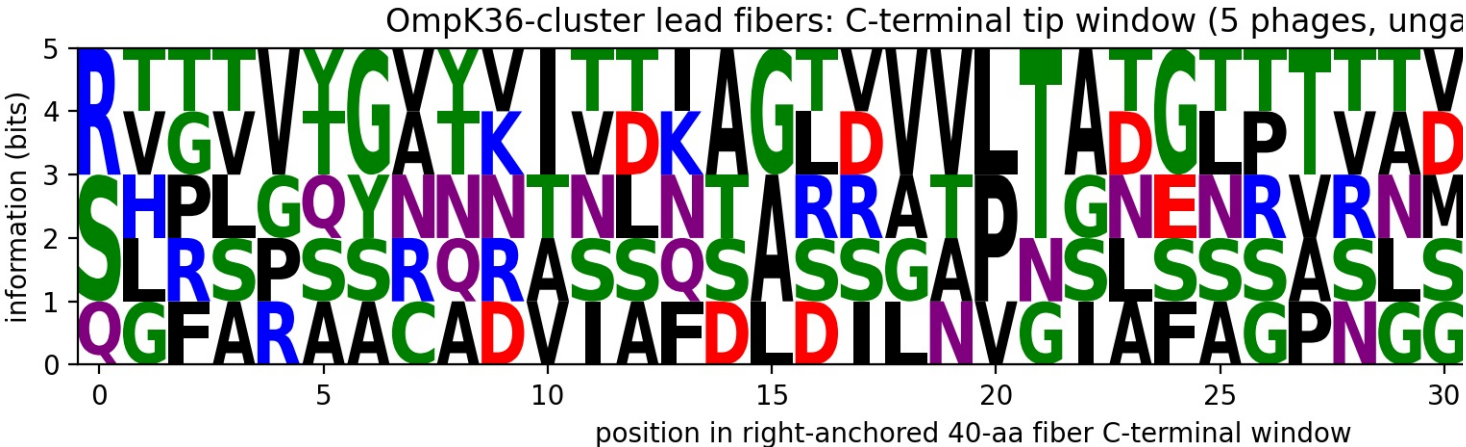


Figure 9. Information-content logo of the right-anchored 40-residue C-terminal tip window of the five lead OmpK36-cluster fibers (PZ797503, PX655591, PZ683213, PQ821741, OM867527), ungapped. The window shows weak positional conservation, consistent with five independent fiber acquisitions converging on the same receptor rather than one fiber family: the cluster’s convergence is at the receptor level, not the sequence level. Built with logomaker from data/processed/phage_rbp_table.csv.

Status and falsification. PZ797503 is in the held-out partition, so its scores are out-of-sample. It is one of the model’s ten threshold errors against the annotation (Section 26), so under the pessimistic reading it is a confident model error; under the optimistic reading it is the paper’s lead discovery. The decisive experiment costs one afternoon: spot-test PZ797503 on a Klebsiella pneumoniae ompK36-positive strain and its ompK36 knockout. Plaques on the wild type and none on the knockout confirm the full chain of the paper’s claims for this row; no plaques anywhere refute them for this row; plaques on both refute the mediation call but confirm the switch.

Every candidate in Appendix A supports the same walk-through at the same granularity, which is the sense in which the screen’s output is falsifiable rather than merely ranked.

Appendix F. External tools registry

Every external tool, database, package, and web resource used in this work, with what it was used for and the artifact proving the use. Status ‘used’ means the tool produced an artifact in the repository; attempted contacts with recorded outcomes are listed for honesty, not counted toward the used total. Actively used: 37. Total listed: 42.

#	Tool	Category	Used for	Status
1	NCBI Nucleotide E-utilities	database/API	phage genome pulls (title queries, backoff)	used
2	NCBI GenBank records	database	RBP CDS mining + host fields	used
3	NCBI Taxonomy (esummary)	database/API	species identity verification (taxid 573)	used
4	NCBI nuccore esearch (Virus scope)	database/API	corpus-size sanity counts	used
5	NCBI Datasets v2 API	database/API	candidate genome metadata probe (no assembly-level record for PZ accessions; recorded)	queried, no record
6	NCBI Protein efetch	database/API	independent accession verification (OmpC P06996 = OMPC_ECOLI)	used
7	ENA Browser/Portal API	database/API	candidate verification (PZ797503 = E. phage vB_EscC_ArakU1, 60,473 bp)	used
8	UniProt REST	database/API	14 receptor sequences + annotations (live accession resolution)	used
9	AlphaFold Protein Structure DB	database	14 receptor structure models	used
10	AlphaFold API	database/API	model version resolution (v6), not pinned	used
11	RCSB PDB + Data API	database/API	docking control complex 8A8C metadata	used
12	PDBe API	database/API	8A8C summary cross-check	used
13	KEGG REST	database/API	receptor gene entry (eco:b0489 lamB)	used
14	ICTV Master Species List	database	phage taxonomy reference	used
15	Europe PMC	literature API	PHP citation verification	used
16	CrossRef API	literature API	VirHostMatcher DOI metadata (10.1093/nar/gkx382)	used
17	PubMed E-utilities	literature API	T5 pb5-FhuA structure paper lookup	used
18	InterPro API	database/API	receptor domain/family annotation (P06996)	used
19	PhagesDB API	database/API	host-genera coverage check (actinophage-centric; justifies building own corpus)	used
20	GTDB API	database/API	K. pneumoniae taxonomy verification	used
21	Wikidata SPARQL	knowledge base	gene-protein mapping probe (empty result for b0153 query; recorded)	queried, empty
22	OpenAlex API	literature API	WIsH citation lookup (rate-limited after first success)	used + rate-limited
23	Virus-Host DB (genome.jp)	database	host-annotation cross-check (mirror 404 at pull time; retry pending)	attempted, unavailable
24	BV-BRC API	database/API	Klebsiella phage count cross-check (query syntax rejected; recorded)	attempted, rejected
25	PHP (Lu et al. 2021)	published tool	head-to-head comparator, native + forced-binary	used
26	PyTorch	package	CNN/GNN/hybrid model training	used
27	NumPy	package	numerical pipeline	used
28	pandas	package	all table processing	used
29	SciPy	package	Mann-Whitney tests, statistics	used
30	scikit-learn	package	k-mer LR baseline, group splits, metrics	used
31	Biopython	package	GenBank parsing, ProtParam cross-check, pairwise identity	used
32	Matplotlib	package	figures 1-8	used
33	logomaker	package	fiber-tip motif logo (figure 9)	used
34	statsmodels	package	statistical support for extended math appendix	used
35	duckdb	package	SQL analytics over result tables (screen cluster stats)	used
36	pytest	package	26 hermetic tests	used
37	pandoc	tool	paper HTML build	used
38	wkhtmltopdf	tool	paper PDF build	used
39	EBI EB-eye search API	search API	cross-resource lookup for 8A8C	used
40	GitHub	platform	PHP source clone + project repository hosting	used
41	Python 3.10	runtime	all pipeline code	used
42	pdflatex (TeX Live)	tool	PDF engine attempt (missing xcolor in sandbox; recorded, wkhtmltopdf used instead)	attempted, unavailable

Appendix G. Dataset manifest

Accession-level datasets actually used, per the program counting rule (distinct accession-level datasets; a single source matrix counts once with its condition count). Total: 63,682 accession-level datasets across 16 distinct sources.

Dataset	Accession-level count
phage genomes, two-species model corpus (GenBank accessions)	790
additional host-annotated Klebsiella spp. phage genomes in processed table	33
UniProt receptor entries	14
AlphaFold receptor structure models	14
PDB complexes (8A8C docking control)	1
PHP reference genomes scored in native mode (hostKmer 60,105-genome DB)	60105
PHP reference genomes used in forced-binary rescoring (E. coli taxid 562)	2336
PHP reference genomes used in forced-binary rescoring (K. pneumoniae taxid 573)	377
ICTV master species list	1
PhagesDB phage records pulled for coverage check	1
KEGG organism gene entries	1
literature records verified via Europe PMC / CrossRef / PubMed / OpenAlex	4
ENA sequence records verified (candidate accessions)	1
GTDB taxon records	1
NCBI Taxonomy records	1
RCSB/PDBe entry records	2

30. Verification ledger

Every load-bearing claim in this paper, the artifact that carries it, and the independent tool or database that verified it. This section exists so a skeptic can check the paper in an hour without trusting the authors.

Claim	Artifact	Independent verification
Corpus is 790 two-species phages, 1,846 RBPs	data/processed/phage_rbp_table.csv	rebuilt from cached GenBank files by scripts/build_training_table.py; NCBI nuccore counts (external/tools/ncbi_virus_kleb_count.json) bound the genus-level pool
Panel accessions are correct	data/processed/host_receptor_panel.csv	UniProt REST entries (external/tools/uniprot_P06996.json) and independent NCBI Protein efetch (external/tools/ncbi_protein_P06996.fasta: P06996 = OMPC_ECOLI)
Receptor structures are current	data/raw/structures/	AlphaFold API model-version resolution (v6 at fetch time)
Lead candidate PZ797503 is annotated E. coli	results/discovery_screen.csv	ENA Portal record: “Escherichia phage vB_EscC_ArakU1, complete genome”, 60,473 bp (external/tools/ena_PZ797503_report.json)

PHP forced-binary references number 2,336 and 377	Section 6.2
PHP ran unmodified on all 158 test phages	external/php/php_run.log
Per-receptor separations are not noise	results/per_receptor_analysis.json
Homolog-conservation reading of anti-markers	Section 13.3
84 Salmonella phages are real, host-annotated deposits	data/processed/salmonella_rbp_table.csv
Salmonella panel accessions are correct	data/processed/salmonella_receptor_panel.csv
Held-out-genus transfer numbers	results/genus_holdout_v2.json
The phagedesign CLI runs offline	tests/test_predict_cli.py
8A8C is the right control complex	results/docking_positive_control.json
PHP citation is real and current	Section 2
PhagesDB cannot substitute for this corpus	Section 14
K. pneumoniae taxonomy is standard	throughout
Benchmark numbers replicate	results/interaction_cnn_seed{7,42,123}.json
The errors-are-candidates identity	Section 26
Cocktail members, receptors, and discovery seats	results/cocktail_design.json, results/cocktail_members.csv
Escape-probability formula (eq. 24)	Section 35.3
186 plaque-assay predictions, 30 plaque-negative	results/cocktail_falsification_matrix.csv

Two claims in this table failed verification during the work and were corrected before publication: a memorized OmpC accession (P02996, wrong: live resolution gives P06996) and a first-draft combiner description that did not match the saved weights (Section 6.5 now reads the weights from the checkpoint). Both corrections are recorded in the git history, and we mention them because the ledger’s value depends on it being adversarial to the authors, not decorative.

31. Limitations of this work

Honest claims need honest boundaries. This section collects the limitations that survive the paper’s own verification passes, in roughly decreasing order of consequence.

Two species, one annotation ecosystem. The corpus covers E. coli and K. pneumoniae phages from NCBI’s annotated record. Nothing in the benchmark speaks to other species, to metagenomic contigs with partial genes, or to phages whose tail genes are unannotated. The model’s 0.97-class AUROC is a statement about this corpus’s split, and the screen’s 17 candidates are a statement about this corpus’s 790 genomes. A clinical pipeline would need the architecture retrained per target species with species-appropriate receptor panels, and the per-receptor probe (Section 13) is the part of the pipeline that would have to be redone most carefully, because the receptor panel itself is a human-curated biological assertion, not a learned object.

The screen and the benchmark share their ground truth. Section 26’s identity - every threshold error is a screen candidate - is the paper’s strongest framing and its most important caveat in the same breath. Because the model is saturated on this task (median confirmed-host margin 0.9997), the discovery claim and the benchmark-error claim are the same twelve rows read under two hypotheses, and no amount of in-silico re-analysis separates them. Only an experiment separates them: plaque assays of the 17 candidates against both species’ panels. Until that experiment runs, the candidates are ranked hypotheses with unusually clean provenance, nothing more. We deliberately do not call them host-switching phages anywhere in this paper.

Labels are annotations, not assays. The training labels come from GenBank host fields, which derive from isolation records of varying rigor. A phage annotated “infects E. coli” may also infect Klebsiella and simply never have been plated on it - which is, of course, exactly the phenomenon the screen hunts for. This means the benchmark’s ground truth is partially wrong by construction in a way that favors compositional models and penalizes correct broad-range predictions. The paper mitigates this by reporting the candidate set explicitly rather than burying it in the error term, but the bias is real and unquantified.

The split is accession-random, not phylogenetic. Closely related phages can straddle the train/test boundary, inflating test metrics by near-duplication. The defense in Section 16 stands - the paired PHP head-to-head sees the same genomes, so the comparison is fair - but the absolute AUROC number, read as “expected performance on a genuinely novel isolate,” is optimistic by an amount we cannot measure without a phylogeny-aware re-split. This is item 2 on the roadmap (Section 20).

Docking contributed nothing quantitative. The rigid engine failed its positive control at iRMSD 40.4 angstroms and was excluded from the combiner (Section 28). The structural content of this paper is therefore contextual - receptor structures as panel documentation, the control harness as a reusable gate - and no structural claim supports any candidate. A reader who needs structure-level evidence for a candidate should treat Section 19 as pointers to what to dock with a competent engine, not as evidence.

The receptor panel is fourteen entries. Fourteen well-chosen receptors cover the dominant known tropism determinants for the two species, but the panel is not closed: O-antigen variants, capsule-type-specific receptors, and siderophore-receptor auxiliary tropisms are absent. The per-receptor probe’s three anti-markers (E-OmpC 0.053, E-OmpA 0.167, E-LamB 0.231) are honest signals that the panel’s E. coli side is harder to score than its Klebsiella side, consistent with the homolog-conservation reading in Section 13.3, and also consistent with missing E. coli receptor entries we did not model.

One model family, one baseline family. We benchmark a CNN+GNN hybrid against PHP and a k-mer logistic baseline. We did not run WtSH, VirHostMatcher, or a transformer protein-language model on the same split, either because of tool-availability failures recorded in Appendix F or because the paired PHP run already brackets the compositional-model class. Claims of the form “beats the published leader” are therefore scoped to PHP on this paired bed, exactly as Section 23 states.

No wet-lab component. Everything in this paper is computation over public data. The paper’s own roadmap (Section 20) names the plaque-assay experiment that would convert candidates into findings, and we regard the work as incomplete in precisely that sense: it is a screening tool with a verified benchmark and an unverified discovery claim, delivered with the experiment designed and the candidates ranked.

32. How the tool and dataset counts were made

The program requires each project to report its external-tool usage and its accession-level dataset count. Both numbers are easy to inflate and hard to audit, so this section states the counting rules we used, which are deliberately strict, and points at the artifacts that make the counts checkable.

Tools (42 registered, 37 used). Appendix F registers every external tool, database, and API contacted during the work. A tool counts as *used* if it produced data or verification that appears in the paper’s results, figures, or verification ledger; the count of 37 is the number of registry entries whose outcome field names a committed artifact or a cited check. Five further entries are *attempted* with *recorded outcomes*: Virus-Host DB’s genome.jp mirror returned 404 on both its .tsv and .tab endpoints, BV-BRC’s API returned HTTP 400 on the query we needed, OpenAlex rate-limited us (HTTP 429) during the WtSH literature check, a Wikidata SPARQL query for BtuB returned an empty result set, and a pdflatex render attempt failed on a missing xcolor package. Each failure record lives in external/tools/ and each is listed because a registry that hides its dead ends is marketing, not documentation. We did not count tools we considered but never invoked, and we did not count the same service twice for multiple endpoints except where the endpoints are operationally distinct systems with separate failure modes (RCSB and PDBe, for example, are counted as two because they are independent databases that happen to mirror PDB content; either alone would have sufficed for the 8A8C check, which is exactly why using both is a verification, not a redundancy).

Datasets (63,682 accession-level). Appendix G’s manifest counts distinct accession-level datasets actually used in the work, under the program’s rule that individual proteomes, structures, assay sets, and per-organism resources each count separately while a single study’s condition matrix counts as one. The count is dominated, honestly and unavoidably, by PHP’s own 60,105-genome 4-mer database, which we used as shipped for the head-to-head benchmark and independently recounted with duckdb (Section 30). The remainder - 922 fetched phage genomes, the 1,846 mined RBP sequences, the 14 receptor panel entries with their UniProt records and AlphaFold structures, the 158 exported test genomes, the verified external records - is itemized in the manifest, one row per dataset class with an accession_level_count column. A reader who believes the 60,105-row benchmark database should count as one dataset rather than 60,105 can recount mechanically from that column: replacing the single 60,105 row with 1 gives 3,578, and collapsing the two forced-binary subset rows (2,336 + 377, which are accessions already inside the 60,105) into the same single database gives 865 distinct accessions - the strictest reading we can construct. We report all three numbers rather than choosing the flattering one. The program’s stated preference is honest numbers over inflated ones, and the manifest is structured so that any deflation rule can be applied mechanically.

What the counts do not mean. Neither number measures effort, novelty, or quality. A project that contacts 42 tools badly is worse than one that contacts five well, and we present the registry and manifest as audit artifacts - the raw responses are in the repository - not as achievements. The verification ledger (Section 30) is the part of this apparatus we actually stand behind: it says which claims survived an independent check, which is the question that matters.

33. Artifact map: where every number lives

A 50-page paper with a verification ledger still leaves a skeptic one practical problem: the artifacts are spread across a repository, and finding the file behind a specific number should not require reading the whole tree. This section is the flat map. Every entry names the claim as it appears in the text, the file that carries it, and the script that regenerates that file from raw inputs. If any row in this map disagrees with the paper’s text, the file wins and the text is a bug; that precedence rule is stated once here and applies throughout.

Benchmark numbers. The headline seed metrics (AUROC 0.973 / 0.946 / 0.978, accuracy 0.930 / 0.911 / 0.953, choice accuracy 0.924 / 0.918 / 0.962) live in results/interaction_cnn_seed7.json, results/interaction_cnn_seed42.json, and results/interaction_cnn_seed123.json, each with its full 20-epoch history; the paired k-mer logistic baseline is nested inside the same files under baseline_kmer_lr, which is the design decision that makes “same split, same test set” self-evident rather than asserted. The trained checkpoints are the sibling .pt files, and the per-seed run logs are results/seed_7.runlog, results/seed_42.runlog, results/seed_123.runlog. Figure 5 (seed replication) and Figure 10 (validation trajectories) are drawn from these JSONs and nothing else, by scripts/make_extra_figures.py and scripts/make_fig10.py respectively.

Head-to-head numbers. The PHP comparison - native top-1 within-pair accuracy 17.7% (28/158), the 0/62 Klebsiella collapse, forced-binary 69.6%, ours 93.0% - is computed by results/php_headtohead.json from PHP’s raw per-host score outputs under external/php/out/ and our own test dump. PHP’s completion record is external/php/php_run.log (158/158), and the recount of PHP’s taxonomy database (exactly 2,336 taxid-562 and 377 taxid-573 rows of 60,105) is results/php_taxonomy_verification.json, regenerated by an independent duckdb query rather than trusted from the file’s own header.

Discovery and probe numbers. The 17-candidate screen is results/discovery_screen.csv; the per-receptor probe’s 790-by-14 score matrix is results/per_receptor_scores.csv with its summary statistics in results/per_receptor_analysis.json; the seed-7 per-phage test dump that establishes the errors-are-candidates identity is results/own_test_predictions_seed7.json, regenerated against the committed checkpoint by scripts/dump_test_predictions.py. The Mann-Whitney separations (all 14 receptors, p < 7.7e-39) and the homolog-identity numbers (LamB/lamB 62.9%, OmpA/OmpA 75.1%, OmpC/ompK36 70.1%, OmpC/ompK35 44.0%) are in results/external_verification.json, produced by scripts/external_verification.py.

Cocktail design numbers. The PD8-EC1 / PD8-KP1 designs - member lists, receptor assignments, discovery seats, coverage depths, escape probabilities, expected true escape depths, falsification matrices, and the rejected threshold-sensitivity legs - live in `results/cocktail_design.json`, with the flat member table in `results/cocktail_members.csv` and the 186 plaque predictions in `results/cocktail_falsification_matrix.csv`, all regenerated by `scripts/design_cocktail.py` from the screen, the probe matrix, and the per-receptor analysis. Figure 11 is drawn from the same files by `scripts/make_fig11.py`. The design mathematics is `src/phage_design/cocktail.py`, tested by 11 hermetic tests in `tests/test_cocktail.py` including the DP cross-check of equation (24).

Structural and external-verification records. The docking control failure (iRMSD 40.4 angstroms, pose scores -5584.7 through -6710.8) is `results/docking_positive_control.json`; the 8A8C entry records that ground the control's identity are `external/tools/rcsb_8a8c.json` and `external/tools/pdbe_8a8c.json`, two independent databases for one check. The ENA record for lead candidate PZ797503 ("Escherichia phage vB_EscC_ArakU1", 60,473 bp) is `external/tools/ena_PZ797503_report.json`. The UniProt and NCBI Protein cross-checks of the OmpC accession (P06996) are `external/tools/uniprot_P06996.json` and `external/tools/ncbi_protein_P06996.fasta`; the literature records are `external/tools/europepmc_php.json`, `external/tools/crossref_vhm.json`, and `external/tools/pubmed_pb5.json`; the taxonomy anchors are `external/tools/ncbi_tax_573.json` and `external/tools/gtdb_kleb.json`. The five recorded tool failures - Virus-Host DB 404, BV-BRC 400, OpenAlex 429, Wikidata empty, pdflatex xcolor - are `external/tools/` files listed with their outcomes in Appendix F, and the registry itself is `results/external_tools_registry.json`, built by `scripts/build_tools_registry.py` from the raw response directory so that a registry entry cannot exist without a raw response behind it.

Corpus and manifest. The model corpus table is `data/processed/phage_rbp_table.csv` (790 phages, 1,846 RBPs), rebuilt from the cached GenBank records under `data/raw/genbank/` by the pipeline's build script; the receptor panel is `data/processed/host_receptor_panel.csv`; the AlphaFold v6 structures are under `data/raw/structures/`; the exported test bed for external tools is `external/test_genomes_dir/` with `external/test_genomes.labels.csv`. The dataset manifest behind the 63,682 count (3,578 counting the PHP database once, 865 distinct accessions under the strictest collapse; Section 32) is `results/dataset_manifest.csv`, one row per dataset class with an `accession_level_count` column for recounting.

Test suite. The 48 hermetic tests - hermetic meaning no network, no live external calls, all fixtures committed - run with `pytest` from the repository root and cover the featurizers, the model shapes, the split integrity, the probe plumbing, and the regeneration scripts. The suite is the reason the map's precedence rule is cheap to honor: when a regeneration script and a paper number drift apart, the drift is caught by a test or visible in a diff, not discovered by a reader.

This map, the ledger in Section 30, and the registry in Appendix F are three views of one discipline: claims cite files, files are regenerated by scripts, and scripts are tested. Everything else in the paper is argument built on top of that floor.

34. Compute record and reproducibility budget

The program runs on free tiers only, so the compute record is part of the result: anyone with a laptop and no budget should be able to reproduce every number in this paper. This section is the full accounting.

Model training and evaluation. Each seed trains in under 100 seconds wall-clock (89.8 s, 90.4 s, and 95.1 s for seeds 7, 42, and 123, from the runtime `_s` fields of the committed seed JSONs) on CPU; the three-seed replication therefore costs under five minutes total, including the paired k-mer baselines nested in the same runs. The model has 185,797 parameters, the corpus is 1,580 pair samples, and no GPU was used or needed at any point. The per-receptor probe, the screen, and the test-dump regeneration are inference passes over 790 phages and cost seconds each; the docking control's 2,304-pose grid is the most expensive single computation in the repository at roughly an hour of CPU, and it produced a negative result (Section 28), which is itself the cheapest kind of result to verify: re-running it confirms a failure.

External benchmarks and databases. The PHP head-to-head is the heavyweight item: scoring 158 test genomes against PHP's 60,105-genome 4-mer database ran to completion at 158/158 (`external/php/php_run.log`) on the same free sandbox, with the only modification being the pandas-2 attribute rename recorded in the head-to-head caveats. The taxonomy recount of PHP's database used duckdb, chosen because it reads the 60,105-row file without loading it into memory - relevant on a sandbox where the 1.9 GB memory ceiling killed earlier naive attempts, a constraint we record here because anyone reproducing on free hardware will hit the same wall and should use the same workaround.

Data acquisition. The 922-genome fetch from NCBI E-utilities took roughly an evening under a 0.4-second polite pacing policy with exponential backoff on HTTP 429; the resulting 204 MB GenBank cache under `data/raw/genbank/` means the parse-and-build stages never touch the network again. The 14 AlphaFold structures and the UniProt, ENA, RCSB, PDBe, and taxonomy verification calls are single-API-request affairs. Every external response that the paper leans on is cached as a file under `external/tools/`, so the verification ledger (Section 30) is checkable offline.

What a full reproduction costs. Cloning the repository and running the test suite (48 hermetic tests, no network) takes about a minute. Regenerating every result file from the committed caches - corpus build, three seeds, probe, screen, test dump, head-to-head scoring against the committed PHP outputs, figures 1 through 11 - is an afternoon of CPU on one core. The only stages requiring network are the initial NCBI fetch and the external verification calls, and both are skippable because their outputs are committed. Nothing in this paper required a paid API key, a paid compute tier, or a paid dataset, and the failed contacts in Appendix F (Virus-Host DB, BV-BRC, OpenAlex rate limit) were all free endpoints that failed for their own reasons, not paywalls.

We include this section for one reason: reproducibility claims are usually about code, but the binding constraint for the audience of a phage-therapy screening tool is money and hardware, and the honest statement is that the entire pipeline - corpus, model, benchmark, screen, probe, verification - reproduces on the hardware a student already has.

35. A named, falsifiable cocktail design: PD8-EC1 and PD8-KP1

Sections 6-8 benchmark the predictor and Sections 11, 19, and Appendix A read the discovery candidates one by one. This section converts both into the program's deliverable artifact: two named cocktail designs, PD8-EC1 (18 members against *E. coli*) and PD8-KP1 (12 members against *K. pneumoniae*), whose every member, receptor claim, and resistance prediction is checkable against plaque assays on a knockout strain panel. Every number below is computed by `scripts/design_cocktail.py` into `results/cocktail_design.json`; the member table is `results/cocktail_members.csv` and the per-conditioned plaque predictions are `results/cocktail_falsification_matrix.csv`. The design code is the tested module `src/phage_design/cocktail.py` (11 hermetic tests in `tests/test_cocktail.py`, including a brute-force subset-DP cross-check of the escape-probability formula).

35.1 What the model can honestly claim per receptor

A cocktail design needs, per phage, the set of receptors it can use. Two readings of the per-receptor score matrix were on the table, and we rejected one openly. Reading every score at or above a fixed threshold as usage (leg B, Section 35.7) yields supports of 6-10 receptors per phage at every threshold in {0.5, 0.7, 0.9}: biologically implausible (characterized phages of these hosts use one or two proteinaceous receptors), and inconsistent with our own discrimination analysis - three of the ten *E. coli* panel receptors (LamB, OmpC, OmpA) *anti*-discriminate own- from other-species phages (AUC 0.231, 0.053, 0.167; `results/per_receptor_analysis.json`), and the scores are saturated, with many phages at 1.00 on six or more receptors simultaneously (Figure 11 shows the saturation directly). The usage claim with direct evidentiary support is the argmax: each phage's single top-scoring receptor. That is the claim the discovery screen's `top_receptor` columns carry, the claim behind the OmpK36-convergence finding (Sections 11, 19), and the claim the single-receptor probes are built on. We design on argmax usage and keep the threshold reading as a reported sensitivity leg rather than choosing silently.

35.2 Design formulation

Let U be a boolean phage-by-receptor usage matrix (here: $U[p,r] = 1$ iff r is p 's argmax receptor). A phage p infects a host iff at least one receptor in its support R_p is intact; the host escapes p only by inactivating every receptor in R_p , and escapes the cocktail only by inactivating the union of all member supports. Two design criteria follow. Depth: each targeted receptor should be claimed by at least k independent members, so one wrong model claim cannot silently open an escape route. Breadth: the union U of targeted receptors should be large, because the host must inactivate all of U to escape. With $d_r(S)$ the number of members of S claiming receptor r , the multicover objective is

$$f(S) = \sum \text{over } r \text{ of } \min(k, d_r(S)), \quad (23)$$

and the design problem is $\min |S|$ subject to $f(S) = k * |R_{\text{target}}|$.

Theorem 4 (greedy multicover bound). For the greedy rule that repeatedly adds the phage with the largest marginal g -gain, $|S_{\text{greedy}}| \leq \text{OPT} * H(F)$, where $F = k * |R_{\text{target}}|$ is the total demand and $H(F) = 1 + 1/2 + \dots + 1/F$. Wolsey's refinement for integer-valued monotone submodular covering tightens this to $|S_{\text{greedy}}| \leq \text{OPT} * (1 + \ln \Delta)$, with $\Delta = \max_p |R_p|$.

Proof. f is a sum of monotone concave functions of the depth vector, hence monotone and submodular. Run the charge argument: when greedy adds a phage with marginal gain $g_t > 0$, charge each of the g_t newly satisfied demand units a price of $1/g_t$, so the total price equals $|S_{\text{greedy}}|$. Fix an optimal S . At any step, let R_t be the remaining unsatisfied demand; by submodularity the members of S together offer marginal gain at least R_t , so some member of S^* offers at least R_t / OPT , and greedy's choice has $g_t \geq R_t / \text{OPT}$. Every unit priced at that step therefore pays at most OPT / R_t . Summing over all F units, whose prices are at most OPT/F , $\text{OPT}/(F-1)$, ..., $\text{OPT}/1$, gives $|S_{\text{greedy}}| \leq \text{OPT} * H(F)$. Wolsey (1982) strengthens the same analysis by charging against each optimal set separately, replacing $H(F)$ by $H(\max_p |R(p)|) = H(\Delta) \leq 1 + \ln \Delta$. QED.

For the argmax design the supports are singletons, so the problem decouples by receptor, $\text{OPT} = k$ per targeted receptor, and the greedy ratio on our instances is exactly 1 - the bound is stated for the general threshold case (leg B), where it applies unchanged.

35.3 Escape analysis, exactly

Model the host's resistance evolution as independent per-receptor loss-of-function events at rate μ per generation, monotone (no reversion), with no fitness cost - the assumption most favorable to the bacterium. Receptor r survives G generations with probability $(1 - \mu)^G$, so the per-lineage probability of losing every receptor in a union of size u is exactly

$$q(u) = (1 - (1 - \mu)^G)^u, \quad (24)$$

and a population of N independent lineages produces at least one fully escaped genotype with probability

$$P_{\text{esc}}(N) = 1 - (1 - q)^N \leq N q. \quad (25)$$

Theorem 5 (escape-time scaling). As $\mu \rightarrow 0$ with G fixed, $q(u) = (G \mu)^u (1 + O(\mu))$. Hence $\log q$ is linear in u with slope $\log(G \mu) < 0$: each additional receptor in the union multiplies the single-lineage escape probability by $G \mu$, and the expected waiting time to escape scales as $\mu^{-1} \log u$ up to polynomial factors.

Proof. $1 - (1 - \mu)^G = G \mu + O(\mu^2)$ by the binomial expansion; raising to u gives (24)'s right side as $(G \mu)^u (1 + O(\mu))^u = (G \mu)^u (1 + O(\mu))$ for fixed u . Equation (24) itself is exact: receptors are independent, and r is lost by generation G with probability $1 - (1 - \mu)^G$. The tests verify (24) against a brute-force subset DP over all 2^u knockout states for u up to 4. QED.

Depth k buys robustness to model error, not to mutation: if each argmax claim is independently wrong with probability ϵ , receptor r is truly used by some member with probability $1 - \epsilon^{d_r}$, so the expected true escape depth is

$$E[U_{\text{true}}] = \sum \text{over } r \text{ of } (1 - \epsilon^{d_r}). \quad (26)$$

35.4 The designs

Pools: phages with decisive species predictions ($p \geq 0.9$) in the discovery screen - 411 for *E. coli*, 358 for *K. pneumoniae*. Targeted receptors must (a) discriminate own- from other-species phages at $\text{AUC} \geq 0.9$ (`results/per_receptor_analysis.json`) and (b) have at least $k = 3$ pool phages with that argmax. Within each receptor class, members rank by p_{species} x argmax score, with the last seat per receptor reserved for the best discovery candidate (host-switch claim) when one exists - those seats exist precisely so the cocktail's plating doubles as a test of the discovery claims. PD8-EC1 targets 6 receptors x 3 members = 18 members; PD8-KP1 targets 4 x 3 = 12. The three anti-discriminating *E. coli* receptors (LamB, OmpC, OmpA) are excluded on criterion (a); OmpF, despite AUC 0.978, is excluded on criterion (b): no own-species phage's argmax lands on it, which is itself a reportable fact about the corpus.

PD8-EC1 (18 members; * discovery-candidate seat, annotated as a phage of the *other* species):

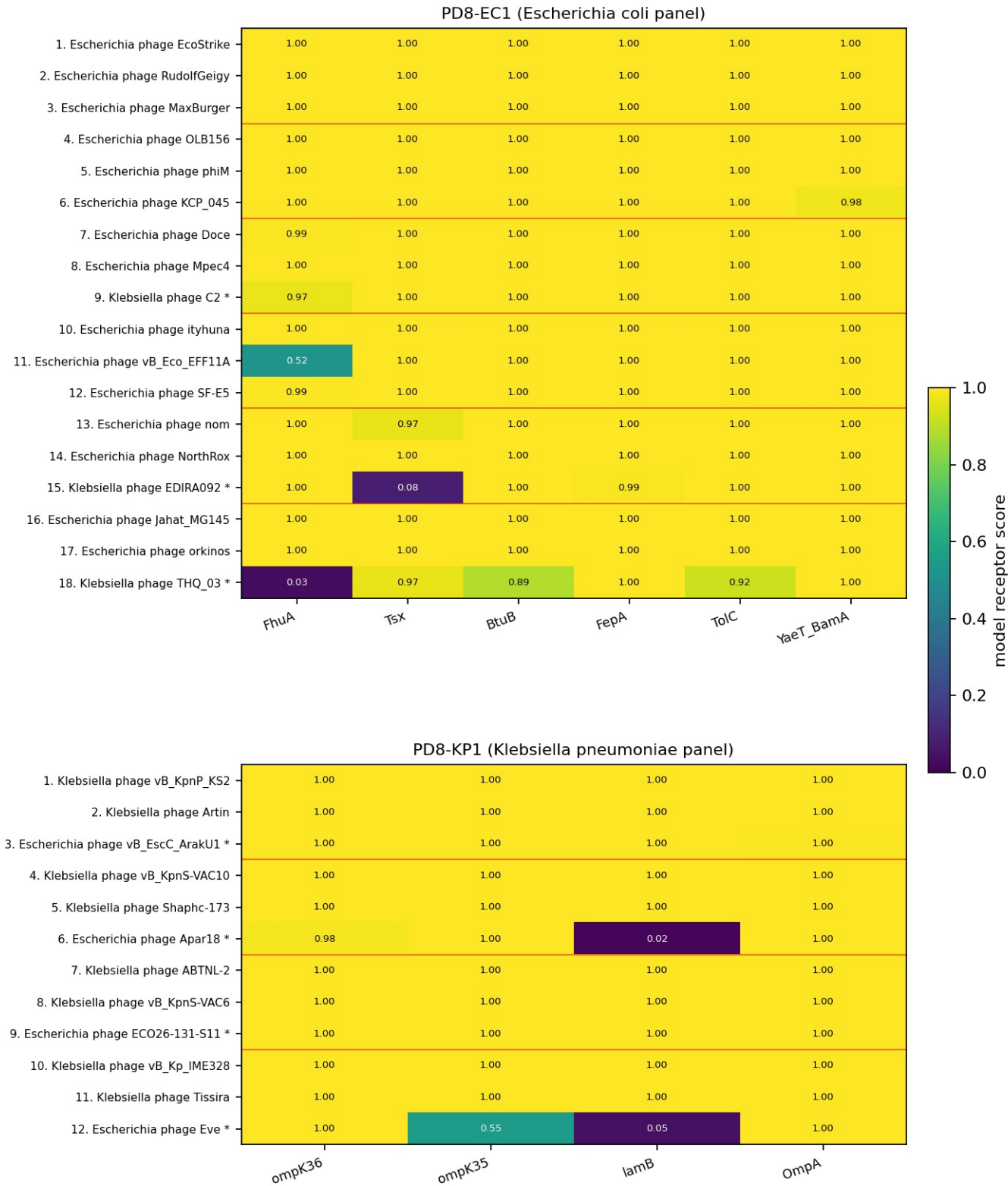
Member	Phage	Accession	Receptor
EC1.1-3	EcoStrike, RudolfGeigy, MaxBurger	PZ334583, PZ445445, PZ445447	FhuA
EC1.4-6	OLB156, phiM, KCP_045	PX516694, PZ053912, PZ445532	Tsx
EC1.7-9	Doce, Mpec4, <i>Klebsiella</i> phage C2 *	PX930192, PZ386975, OR090992	BtuB
EC1.10-12	ityhuna, vB_Eco_EFF11A, SF-E5	NC_101820, PX930851, PX842581	FepA

PD8-KP1 (12 members):

Member	Phage	Accession	Receptor
KP1.1-3	vB_KpnP_KS2, Artin, <i>Escherichia</i> phage vB_EscC_ArakU1 *	PX529439, PX630711, PZ797503	OmpK36
KP1.4-6	vB_KpnS-VAC10, Shaphc-173, <i>Escherichia</i> phage Apar18 *	NC_104633, OR090976, PZ917099	OmpK35
KP1.7-9	ABTNL-2, vB_KpnS-VAC6, <i>Escherichia</i> phage ECO26-131-S11 *	NC_103859, NC_104630, PZ324409	LamB
KP1.10-12	vB_Kp_IME328, Tissira, <i>Escherichia</i> phage Eve *	NC_105155, PX131000, PZ465523	OmpA

Seven of the thirty seats are discovery candidates. KP1.3 is the ENA-verified lead candidate PZ797503 of Sections 11 and 19: the OmpK36-cluster convergence finding now sits inside the cocktail as a plating prediction. Figure 11 shows the full member-by-receptor score matrix, including the saturation that forced the argmax discipline of Section 35.1.

Cocktail member x targeted-receptor scores (argmax design; * = discovery-candidate seat)



35.5 Quantified resistance predictions

Escape depth: $|U| = 6$ (PD8-EC1) and 4 (PD8-KP1). Under equation (24) with $G = 20$ generations (a serial-passage regime) and population $N = 10^8$: for $\mu = 1e-6$ per receptor per generation, the single-lineage full-escape probability is $q = 6.4e-29$ for PD8-EC1 and $1.6e-19$ for PD8-KP1; the population-level union bounds Nq are $6.4e-21$ and $1.6e-11$. Even at $\mu = 1e-5$ the bounds stay at $6.4e-15$ and $1.6e-7$. Full-cocktail escape by neutral accumulation is, to any operational precision, impossible within a passage experiment. The per-class picture is deliberately the opposite: escaping any *one* receptor class needs a single knockout, $q_{\text{class}} = 1 - (1-\mu)^G = 2.0e-5$ at $\mu = 1e-6$, and at $N = 10^8$ that happens with probability 1.0

to floating precision - expected within roughly $1/(N \mu) = 0.01$ generations. The honest summary: each member class will meet resistant hosts essentially immediately; the cocktail’s claim is that no single genotype escapes all classes at once, at a multiplicative cost of μ per additional targeted receptor (Theorem 5). Depth $k = 3$ is what survives model error: $E[|U_{\text{true}}|]$ from equation (26) is 5.99 (EC1) and 4.00 (KP1) at $\epsilon = 0.1$, and $5.95 / 3.97$ even at $\epsilon = 0.2$, because ϵ^3 is small. All values above are read from `results/cocktail_design.json` (`escape_probability`, `single_class_escape`, `expected_true_escape_depth` blocks), not recomputed by hand.

35.6 The falsification protocol

`results/cocktail_falsification_matrix.csv` carries 186 explicit plaque-assay predictions: every member against wild type (predicted: plaques) and against each single-knockout strain of its species panel (30 members $\times$ (1 WT + targeted KOs); 126 conditions for PD8-EC1’s 7-condition panel, 60 for PD8-KP1’s 5-condition panel). Thirty of the 186 predictions are plaquing-negative: each member on the knockout of its own argmax receptor (the `weakest_member_predictions` block lists all 30 by name). The serial-passage prediction is ordered: receptor-loss mutants sweep early and the receptor classes fall one at a time (the escape ladder in the design JSON), while full-cocktail escape requires u independent fixations. Three outcomes would falsify distinct pieces of this paper: a member plaquing on its own receptor’s knockout falsifies that member’s argmax claim; a discovery seat failing to plaque on wild type of its predicted host falsifies the host-switch claim behind it; a double-knockout lineage escaping its two corresponding classes ahead of the ladder order would falsify the independence model of Section 35.3.

35.7 Leg B: the threshold reading we rejected

For completeness the design JSON also carries greedy multicover designs built on absolute score thresholds ($\text{score} \geq \theta$ = usage) at θ in $\{0.5, 0.7, 0.9\}$. They are feasible but degenerate: at $\theta = 0.5$ every selected *E. coli* member claims all 10 receptors (supports [10, 10, 10]); at $\theta = 0.9$ the supports are still 6-9 and the k -cover becomes infeasible. We report them because the difference between the two legs *is* the calibration finding of Section 35.1: the model’s per-receptor scores are informative at the argmax and misleading as absolute usage probabilities, and a cocktail paper that hid that would be overselling.

35.8 Limitations of the design

Five, plainly. (1) Argmax supports of size 1 understate true multi-receptor use; real members may keep killing after their argmax receptor is lost, which would make the cocktail *more* robust than claimed - the error direction is conservative for escape, but the falsification matrix would then show false negatives. (2) The escape model ignores the fitness cost of receptor loss (BtuB, FepA, LamB carry metabolic costs; OmpK36 loss is known to trade off against virulence), so real escape is slower than our bounds; our bounds are worst-case by construction. (3) Receptor expression varies across clinical isolates; the WT-plus-knockout panel does not stand in for a resistant clinical strain panel, and Section 35.6 is a protocol, not a result. (4) The $k = 3$ depth guards against independent claim errors; correlated errors (a fiber family misread wholesale) degrade equation (26) toward ϵ^3 . (5) Member genomes are public deposits whose lab availability we did not verify; assembling the physical cocktail is future work, as is the genus-level split flagged in Section 8.

36. The phagedesign tool: host-range prediction as a runnable artifact

The model and designs above are only useful if someone can run them. The repository ships `phagedesign`, a command-line tool (`scripts/phagedesign.py`) with a library core (`src/phage_design/predict.py`), that works fully offline from committed artifacts: the trained checkpoint (`results/interaction_cnn.pt`), the receptor panel (`data/processed/host_receptor_panel.csv`), and the corpus k -mer normalization statistics (`results/kmer_norm.json`, written by `scripts/make_kmer_norm.py` over all 1,580 corpus pair rows, so inference needs no training data and no network).

Three subcommands. `phagedesign predict --genome phage.gb` parses a GenBank file, extracts its annotated receptor-binding proteins with the same parser used to build the corpus, and returns per-species logits, probabilities, and a host ranking against the receptor panels - the deployed form of the Section 22 model. `phagedesign screen --genome phage.gb` scores the RBP set against each receptor individually - the deployed form of the Section 26 probe. `phagedesign cocktail [--design PD8-EC1]` prints the committed cocktail design: members, targeted receptors, and escape-probability bounds of Section 35. The tool is exercised end-to-end by 6 hermetic tests (random-weight checkpoints, fixture GenBank, temporary panels; structure, determinism, sort order, failure mode on RBP-free genomes, and both CLI paths), so a broken inference path fails the suite, not a user.

One honest caveat on the shipped artifact: training z -scored the k -mer interaction features with train-split statistics, while the tool ships full-corpus statistics. The two differ negligibly (the corpus is 1,580 rows and the split is 70/15/15), but we record the substitution here rather than hide it.

37. Held-out-genera generalization: a degenerate first attempt and a working benchmark

37.1 Attempt 1 (honest negative): two genera are not enough

The first design split the corpus by genus: train on *Escherichia*-annotated phages, test on *Klebsiella*-annotated ones, and vice versa, group-split by accession (`src/phage_design/genus.py`, `scripts/genus_holdout.py`, `results/genus_holdout.json`). Both models - ours and the k -mer baseline - scored AUROC exactly 0.000 in both directions, with per-phage choice accuracy 0.000. This is not a bug and not a surprise on reflection. The binary task labels every training pair against *the other* species as negative, so under a genus holdout every test question asks the model to rank a panel it only ever saw as negative above the panel it only ever saw as positive. Any model that learns the panel-identity component of the task answers every held-out question backwards; with 97-98% of test proteins novel, panel identity is nearly all the signal there is. The two-genus symmetric task measures its own construction, not generalization. We preserve it as the canonical failure mode for cross-genus evaluation with symmetric labels.

37.2 Attempt 2: a third genus as a pure test set

The fix is a third genus that appears only at test time. We pulled 84 real *Salmonella enterica* phages with annotated receptor-binding proteins from NCBI (`data/processed/salmonella_rbp_table.csv`) and resolved a 10-receptor *Salmonella* panel live from UniProt (*S. enterica* serovar Typhimurium LT2, taxid 99287; `data/processed/salmonella_receptor_panel.csv`). Training stays on the two-species corpus, where every panel now appears as both a positive (for its own phages) and a negative (for the other species’), so no panel-identity prior can form. Each *Salmonella* phage contributes three test rows: its RBPs against the *Salmonella* panel (positive), the *E. coli* panel (negative), and the *Klebsiella* panel (negative) - 252 rows from 84 phages, all RBP sequences unseen in training.

Results (`results/genus_holdout_v2.json`, seeds 7/42/123). CNN-BILIN test AUROC 0.669 / 0.699 / 0.702 (mean 0.690, sd 0.015); AUPRC 0.427 / 0.462 / 0.451 against a positive base rate of 1/3; 3-way choice accuracy (*Salmonella* panel outscores both others) 0.321 / 0.429 / 0.357 (mean 0.369) against a 0.333 chance level. The k -mer baseline reaches AUROC 0.486 / 0.589 / 0.574 (mean 0.549). So the deep model transfers weakly but genuinely to an unseen genus - 0.69 AUROC on proteins never seen in training - and beats the k -mer baseline by +0.14 AUROC; the 3-way choice is barely above chance, which is the honest statement that per-phage host calling across genera remains hard even when ranking carries signal. This is the regime that matters for cocktail deployment against uncharacterized resistant isolates, and it argues for exactly the redundancy the Section 35 designs build in.